

**{110} {111} and {112} {111} dislocation behavior in W-Re alloys**Patrick F. McNutt<sup>1,\*</sup>, Pulkrit Garg<sup>2</sup>, Morgan R. Jones<sup>3</sup>, Tresa M. Pollock<sup>1</sup>, and Irene J. Beyerlein<sup>1,2</sup><sup>1</sup>*Materials Department, University of California Santa Barbara, Santa Barbara, California 93106, USA*<sup>2</sup>*Department of Mechanical Engineering, University of California Santa Barbara, Santa Barbara, California 93106, USA*<sup>3</sup>*Computational Multiscale Department, Sandia National Laboratories, Albuquerque, New Mexico 87123, USA*

(Received 3 June 2026; revised 24 July 2026; accepted 27 August 2026; published 24 September 2026)

Tungsten (W) is a refractory body-centered-cubic (bcc) metal with exceptional high-temperature strength and an extremely high melting point. However, its ductile-to-brittle transition temperature (DBTT) occurs above room temperature, significantly restricting its formability and, consequently, its range of applications. One approach to modifying the plastic behavior of W at low temperatures is alloying it with rhenium (Re), which enhances its ductility, though the underlying dislocation mechanisms responsible for this improvement are not fully understood. In this work, the phase field dislocation dynamics model is extended to treat {112}{111} dislocation dynamics in a bcc crystal and applied to elucidate the effect of Re additions on dislocation mechanisms in W. The model is used to simulate thermally activated glide of long, initially screw and edge dislocations in W and W-20%Re for a range of elevated temperatures that includes their DBTT. We show that for both materials and at all temperatures, the critical resolved shear stress (CRSS) for {112} glide is higher than for {110} glide and the CRSS for screw dislocations is higher than edge dislocations. The CRSS is higher in the {112} antitwinning (AT) direction than {112} twinning (T) direction up to a certain temperature, at which they become nearly equal. The {112}T/AT equivalence temperature is lower for W-20Re than pure W. We find that Re increases the activation enthalpy for screw dislocation glide for the {110} mode, but decreases it for the {112} mode. Together, these findings suggest that Re additions promote slip activity from being primarily {110} slip to both {110} and {112}T/AT slip.

DOI: [10.1103/3csr-l6hc](https://doi.org/10.1103/3csr-l6hc)**I. INTRODUCTION**

Tungsten (W) is a widely used structural material known for its exceptional physical properties, including a high melting point, high thermal conductivity, low coefficient of thermal expansion, and high-temperature creep resistance [1]. However, its brittleness at room temperature presents a significant drawback, severely limiting both its formability and mechanical performance. This response is common among all group VI metals, e.g., tungsten, molybdenum, and chromium, but has long been recognized as being substantially mitigated through alloying with rhenium (Re) in the case of W [2–4].

Plastic deformation in crystalline materials depends on how easily dislocations glide through the crystal lattice. Therefore, studying the movement and mechanisms of dislocations in a material is crucial for understanding its deformation behavior. In body-centered-cubic (bcc) metals, the shortest lattice vectors are the  $\frac{a}{2}\langle 111 \rangle$  family and these correspond to the glide direction. There are no atomically compact planes and the three possible planes in which the  $\frac{a}{2}\langle 111 \rangle$  glide direction lies are {110}, {112}, and {123}. The most widely spaced planes among the three are the {110} planes and, for this reason, the {110} planes are expected to be the most active or favored planes for slip.

The high Peierls stress for bcc metals is linked to the nonplanar core of the screw-oriented dislocations and the core

of the {110}{111} screw dislocation is the most well studied. According to the Peierls-Nabarro model [5], the core of a screw-oriented full dislocation is spread symmetrically on the three close-packed {110} planes intersecting the {111} dislocation line. The large Peierls barrier for a screw dislocation segment to move one lattice periodicity arises because the fractional dislocations extended on the {110} planes must contract when activated. For a long screw dislocation, only a portion of the line overcomes the Peierls barrier, moves forward  $1b$ , and forms a pair of kinks in the line [6]. These kinks have nonscrew character and a much lower Peierls stress and can migrate along the line and thereby advance the entire dislocation line forward  $1b$ .

Whether dislocation glide on higher-order slip planes, such as the {112}{111} mode, is easier or preferred over the common {110} slip mode is determined by experimental observations or atomistic calculations of Peierls barriers or Peierls stresses. Even so, results vary across studies [7]. Dislocation glide via the {112}{111} mode has been observed in bcc metals, especially at higher temperatures [8]. Unlike {110} glide, {112} glide is asymmetric, with an easier (“twinning”) direction and harder (“antitwinning”) direction, and some studies report that glide in the {112}{111} twinning direction can be easier than in {110} [9]. Screw-type dislocations can also cross slip. It is, therefore, also possible that dislocations gliding on a {112} plane originated via cross slip from a {110} plane where they were first activated [9,10]. Notions of frequent cross-slip events leading to changes from a {110} slip plane to a different {110} or {112} plane have

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led to the idea of “wavy glide” or similar changes among the three modes,  $\{110\}$ ,  $\{112\}$ , and  $\{123\}$ , to the concept of “pencil glide” [11–14]. Understanding which dislocation slip modes are selected and at which temperatures is crucial for understanding the deformation behavior of bcc metals.

The addition of Re to W has long been recognized as one of the most effective alloying strategies for improving its otherwise limited ductility. Early reports dating back to the 1960s showed that even modest Re additions could significantly increase the tensile elongation of W alloys at intermediate temperatures, a phenomenon now known as the “Re effect” [15–17]. Alloying additions of Re up to 25 wt% are beneficial to W alloys as Re can decrease the ductile-to-brittle transition temperature (DBTT) of W, while also increasing its strength retention at high temperature [3,18]. Subsequent studies established that the mechanical behavior is strongly compositionally dependent. For instance, creep strength reaches a maximum at about 6–8 wt% Re [19–22]. Optimum combinations of strength and ductility are evidently achieved for Re content of 3–10 wt%. The concentrations that minimize the DBTT are near the solubility limit (25 wt%) [23]. However, alloying Re into W presents some drawbacks, including higher material costs and a reduction in its weldability [24].

The mechanisms behind the Re-induced ductilization of W alloys have been the subject of longstanding debate, with multiple explanations put forward. The first proposal came from Fleischer *et al.*, who suggested that elastic nonuniformities introduced by Re solute atoms interact with both screw and edge dislocations, leading to solute hardening [25]. Rao *et al.* proposed an alternative mechanism in which solute-dislocation core interactions modify the kink-pair nucleation rate [26]. Predictions from their model agreed with polycrystalline yield data of W-Re alloys, indicating that relative to pure W, Re additions produce softening at low temperatures and strengthening at higher temperatures. Regarding dislocation activation, proposals that explain the effect of Re alloying additions fit into two categories. The first is solid solution softening, described by a reduction in the critical resolved shear stress (CRSS) to move a dislocation caused by Re additions. This response is believed to increase the mobility of  $\frac{a}{2}\langle 111 \rangle$  screw dislocations [27,28]. Both lower CRSS and higher mobility can arguably improve ductility. The second is a shift in the easier slip mode from  $\{110\}$  to  $\{112\}$  with Re additions [15,29]. Some others have proposed the improvement in ductility results from a transition to an asymmetric screw core, which results in dislocations gliding in a zigzag manner on the  $\{112\}$  planes [30,31]. Considering the competing or varying theories posed to date, the role of Re alloying additions in changing dislocation behavior in W remains unclear. The effects of Re on basic relationships, such as  $\{110\}$  vs  $\{112\}$  twinning vs  $\{112\}$  antitwinning mode selection, screw/edge differences, and sensitivity to temperature could help to explain the mechanism for the ductilization provided by Re.

Computational techniques such as molecular dynamics (MD) and density functional theory (DFT) have previously been used to tackle the ductilization question of Re in W by studying single solute/dislocation interactions. It has been reported that alloying W with Re increases the number of

$d$  electrons, which results in the  $\frac{a}{2}\langle 111 \rangle$  screw dislocation core becoming asymmetric, changing the slip plane from  $\{110\}$  and promoting  $\{112\}$  slip when applying an external strain [30]. MD studies have also measured the stress-strain response of W-Re alloys at 300 K and found that Re atoms pin dislocation motion and encourage dislocation reactions [32]. The timescales and length scales of current MD and DFT capabilities are not sufficient to capture thermally activated motion of long dislocations (hundreds of nm). Other modeling techniques, such as kinetic Monte Carlo (kMC) and discrete dislocation dynamics (DDD) studies, are mesoscale models that overcome some of these limitations, but at the sacrifice of atomic-scale fidelity. kMC studies found low concentrations of Re in W soften the Peierls stress of a screw dislocation, while higher concentrations harden it [33]. The study only considered  $\{110\}$  dislocations in W-Re; however, DDD has simulated  $\{110\}$ ,  $\{112\}$ , and  $\{123\}$  slip in the deformation of W but not yet with Re as an alloying addition [34]. Another mesoscale model, phase field dislocation dynamics (PFDD), offers an energy-based framework for studying dislocation behavior over extended timescales and length scales [35,36]. PFDD is especially useful for analyzing dislocation core structures, glide mechanisms, and critical stresses in different metallic materials at finite temperature [37–40]. It predicts the minimum-energy pathways and configurations that dislocations follow within a three-dimensional (3D) single crystal. Recent developments have extended PFDD to use bcc-specific nonorthogonal computational grids, allowing for an approximate representation of the distinct core structures of screw and edge dislocations on  $\{110\}$  planes and enabling the modeling of nonplanar core spreading and cross slip of screw dislocations [36,41]. Langevin dynamics has also been incorporated to extend the PFDD framework to finite temperature and capture thermally activated processes, such as kink-pair nucleation [40–42].

Here, using PFDD, we aim to offer a mesoscale, dislocation-based rationale underlying the effect of Re on the deformation behavior of W. The goal of this work is twofold. First, we extend PFDD to simulate  $\{112\}\langle 111 \rangle$  glide, including capturing screw/edge differences, finite-temperature kinetics, and the twinning/antitwinning asymmetry in glide resistance, based solely on the generalized stacking fault energy curves and elastic moduli from atomic-scale simulation. Second, we apply this framework to investigate the thermally activated glide of dislocations on both  $\{110\}$  and  $\{112\}$  slip modes for a range of elevated temperatures (up to 600 K). Simulations are performed for long, initially straight screw and edge dislocations in both pure W and the W-20at%Re alloy. The alloying amount and temperature ranges studied are motivated by experimental observations where pure W has been observed to show brittle macroscopic plastic response at 600 K and below. However, alloying W with 20 at% Re or higher improves the macroscopic ductility and increases the yield strength by up to 500 MPa over the temperature range of 300–600 K [15]. While the DBTT in W is highly processing dependent, an increase in Re content to near its solid solution limit can lead to significant reductions in DBTT by up to 200 K [21,43,44]. The PFDD predictions suggest Re alloying additions would encourage the use of more slip systems, increasing the number of slip systems from  $\{110\}$  slip

only to {110} and {112} T/AT slip. Specifically, Re increases the activation enthalpy in {110} and substantially decreases it in {112} T/AT and lowers the temperature at which the T/AT asymmetry is removed.

## II. METHODOLOGY

### A. PFDD model with finite temperature

PFDD adopts the phase field formalism and applies it to study the motion of dislocations in crystalline materials [45]. PFDD employs a set of scalar order parameters  $\phi^\alpha(\mathbf{r})$ , each representing the amount of slip at a lattice point  $\mathbf{r} = (x, y, z)$  on slip system  $\alpha$  in a three-dimensional (3D) crystal, where  $\alpha$  ranges from 1 to  $N$ , and  $N$  is the total number of slip systems (order parameters). Unslipped regions of the lattice correspond to  $\phi^\alpha(\mathbf{r}) = 0$  and slipped regions by  $\phi^\alpha(\mathbf{r}) = 1, 2, 3$ , and so on for the number of times the region has been slipped. The slip direction  $s^\alpha$  can be either positive or negative, corresponding to the sign of the Burgers vector.

Order parameters in the PFDD model can be assigned to any number of slip systems and families of slip systems. For the bcc crystal structure, previous work has treated the {110}{111} slip family using three order parameters, one for each of the following slip systems: (110)[ $\bar{1}11$ ], (01 $\bar{1}$ )[ $\bar{1}11$ ], and (101)[ $\bar{1}11$ ] [46,47]. In this way, the model predicts that a [ $\bar{1}11$ ] screw dislocation adopts a nonplanar core structure.

The total energy density  $\psi$  is composed of three contributions: the lattice energy density  $\psi_{\text{latt}}$ , elastic strain energy density  $\psi_{\text{elas}}$ , and the externally applied energy density  $\psi_{\text{ext}}$ ,

$$\psi(\phi) = \psi_{\text{latt}}(\phi) + \psi_{\text{elas}}(\phi) - \psi_{\text{ext}}(\phi). \quad (1)$$

The energy functional usually includes a gradient energy term that corresponds to the penalty associated with the change in order parameter at the dislocation line. It is omitted in this work. In PFDD, the elastic stiffness will regularize the dislocation width and, therefore, the width remains finite even without the gradient term. Moreover, prior PFDD studies have demonstrated that the gradient term produces second-order effects in the structure of the dislocation core [48,49].

The elastic strain energy density is given by

$$\psi_{\text{elas}}(\phi) = \frac{1}{2} C_{ijkl} (\epsilon_{ij} - \epsilon_{ij}^p(\phi)) (\epsilon_{ij} - \epsilon_{ij}^p(\phi)), \quad (2)$$

where  $C_{ijkl}$  is the elastic stiffness tensor of the crystal,  $\epsilon_{ij} = \epsilon_{ij}^e + \epsilon_{ij}^p$  is the total strain, and  $\epsilon_{ij}^e$  is the elastic strain. The plastic strain  $\epsilon_{ij}^p$  is given by

$$\epsilon_{ij}^p(\phi) = \frac{1}{2} \sum_{\alpha=1}^{n_s} \frac{b^\alpha}{d^\alpha} \phi^\alpha (s_i^\alpha n_j^\alpha + s_j^\alpha n_i^\alpha), \quad (3)$$

where  $b^\alpha$  is the Burgers vector magnitude,  $d^\alpha$  is interplanar spacing, and  $n^\alpha$  is the slip plane normal for slip system  $\alpha$ . The external energy density takes the form

$$\psi_{\text{ext}}(\phi) = \sigma_{ij}^{\text{app}} \epsilon_{ij}(\phi), \quad (4)$$

where  $\sigma_{ij}^{\text{app}}$  is the externally applied stress state.

The lattice energy  $\psi_{\text{latt}}$  is the potential energy change associated with shear displacement across a glide plane. For the (110) plane and [ $\bar{1}11$ ] slip direction in the bcc lattice, the

dependence of this potential with  $\phi$  is well described by

$$\psi_{\text{latt}\{110\}\{111\}}(\phi) = \sum_{\alpha} \frac{\gamma_{\text{usf}}^\alpha}{d^\alpha} \sin^2(\pi \phi^\alpha), \quad (5)$$

where the maximum  $\gamma_{\text{usf}}^\alpha$  is the unstable stacking fault energy. The energetic surface or curve for  $\psi_{\text{latt}}$  is usually obtained from an atomic-scale calculation of the generalized stacking fault energy (GSFE) curve for a given crystallographic plane, e.g., the (110) plane.

The stochastic time-dependent Ginzburg-Landau (TDGL) equation is employed to compute the evolution in the set of order parameters  $\phi$  with time,

$$\frac{\partial \phi^\alpha(\mathbf{r})}{\partial t} = -L \frac{\delta \psi}{\delta \phi^\alpha(\mathbf{r})} + \theta(\mathbf{r}, t), \quad (6)$$

where  $L$  is the non-negative TDGL coefficient controlling the rate of relaxation toward equilibrium, and  $\theta$  is a stochastic force term that drives random thermal fluctuations about the equilibrium configuration at finite temperature ( $T$ ) [50]. A complete description of the numerical implementation of Langevin dynamics within the PFDD framework can be found elsewhere [42]. Briefly, the stochastic force term is given by

$$\theta(n, m) = \sqrt{\frac{2k_B T L}{\Omega t}} \rho_i, \quad (7)$$

where  $k_B$  is the Boltzmann constant,  $\Omega$  is the volume element over which the order parameter is defined, and  $\rho\{\rho : \rho_i, i = 1, \dots\}$  is an independent random number drawn from a standard Normal distribution  $N(0, 1)$ . The physical time step  $\Delta t$  is selected to be small enough to resolve thermally activated processes, yet large enough to allow the dislocation to remain in the activated state without interference from the initial activation event. With these considerations, the dimensionless time step  $\Delta \tau$  is of the order of  $10^{-5} \omega/L\mu$ , where  $\omega$  is the Debye frequency and  $\mu$  is the effective isotropic shear modulus. The Debye frequency for W and Re is of the order of  $10^{13}$  Hz (e.g.,  $\omega$  for W  $8.0 \times 10^{12}$  Hz and  $\omega$  for Re  $8.6 \times 10^{12}$  Hz) [51,52]. The time step for convergence of the TDGL is related to the curvature of the free energy with respect to the order parameter [53]. We find that for the present application, this criterion is satisfied when  $\Delta \tau = \frac{L\mu\Delta t d_0}{d} < 0.25$ , where  $d_0$  is a reference interplanar spacing and  $d$  is the interplanar spacing of  $\phi^\alpha$ . In all PFDD simulations that follow,  $\Delta t \sim 10^5/\omega = 10^{-8}$  s and  $d_0$  is the {110} interplanar spacing.

### B. Extension of PFDD to {112}{111} dislocations

In this work, we advance PFDD to account for {112}{111} dislocations, including the nonplanar core of the screw-character dislocations and the twinning/anti-twinning asymmetry in glide resistance. Order parameters  $\phi^\alpha$  associated with slip on a {112}{111}-type slip system  $\alpha$  are established. Without loss of generality, we simulate [ $\bar{1}1\bar{1}$ ] dislocations and therefore assign three order parameters associated with the slip systems sharing this slip direction, which are (112)[ $\bar{1}1\bar{1}$ ], ( $\bar{1}21$ )[ $\bar{1}1\bar{1}$ ], and ( $2\bar{1}1$ )[ $\bar{1}1\bar{1}$ ].

To account for changes in the shape of the GSFE for {112}{111} slip, the lattice energy term  $\psi_{\text{latt}}$  must also be

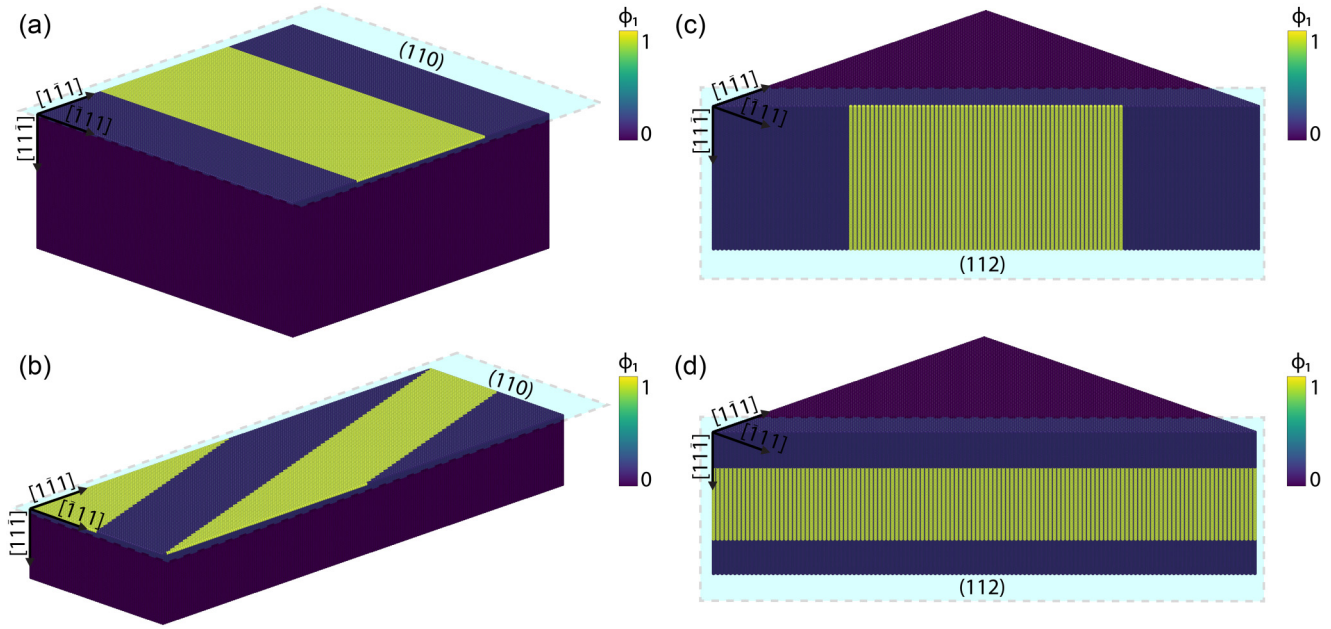


FIG. 1. Initialization of (a) (110) screw dislocation dipole, (b) (110) edge dislocation dipole, (c) (112) screw dislocation dipole, and (d) (112) edge dislocation dipole. The light-blue planes mark the habit plane, either (110) or (112).

updated. Generally, this GSFE is asymmetric and, for this reason, we adopt the following expression for the lattice energy term for the  $\{112\}\{111\}$  slip systems:

$$\psi_{\text{latt}\{112\}\{111\}}(\phi) = \sum_{\alpha} \frac{\gamma_{\text{usf}}^{\alpha}}{d^{\alpha}} \sin^2[\pi(\phi^{\alpha} - c)^B], \quad (8)$$

$$\frac{\partial \psi_{\text{latt}\{112\}\{111\}}}{\partial \phi} = \frac{\gamma_{\text{usf}}^{\alpha}}{d^{\alpha}} \pi B (\phi^{\alpha} - c)^{B-1} \sin[2\pi(\phi^{\alpha} - c)^B], \quad (9)$$

where  $\gamma_{\text{usf}}^{\alpha}$  is the peak in the GSFE, the unstable stacking fault energy, and  $c = \text{floor}(\phi)$ . The exponent  $B$  is related to the deviation in the lattice displacement from  $0.5b$  of the peak  $\gamma_{\text{usf}}^{\alpha}$  from the GSFE, i.e.,

$$B = \frac{\ln(0.5b)}{\ln(\delta_{\text{peak}}^{\alpha})}. \quad (10)$$

The parameter  $B$  can be described as an asymmetry parameter, where the more  $B$  deviates from unity, the more asymmetric the  $\{112\}\{111\}$  GSFE curve. The twinning direction is associated with a GSFE peak location,  $\delta_{\text{peak}}^{\alpha}$ , lying at shift greater than  $0.5b$ , and vice versa for the antitwinning direction.

### C. Simulation initializations

The simulation cell is a nonorthogonal 3D grid with grid points coinciding with bcc lattice points, as illustrated in Fig. 10 of the Appendix. A dislocation dipole, consisting of two dislocations with the same dislocation line but opposite Burgers vectors, is inserted on a plane of the cell. Figure 1 presents the simulation models for edge and screw dipoles placed on the (110) and (112) planes.

As shown in Fig. 1(a), for  $\{110\}$  glide, a screw dislocation dipole can be built by initializing specific grid points on the bcc computational grid to 0 and 1 on the (110) plane, such that

the boundaries between the two regions, which describe the location of the initialized dislocation, are along the direction of the Burgers vector,  $[\bar{1}11]$ . Similarly, for the edge dislocation case of  $\{110\}$  glide, the dislocation is initialized with its line direction perpendicular to the Burgers vector, along the  $[1\bar{1}2]$  direction in the (110) plane, seen in Fig. 1(b).

To study glide on the  $\{112\}$  planes, a similar procedure is performed. In the (112) plane, however, the two orthogonal directions of  $[11\bar{1}]$  and  $[1\bar{1}0]$  can be used to describe adjacent grid points on the bcc computational grid. As shown in Fig. 1(c), to initialize a screw dislocation dipole for  $\{112\}$  glide, the dislocation is placed on the (112) plane and along the direction of the Burgers vector  $[11\bar{1}]$ . However, for the edge dislocation dipole case for  $\{112\}$  glide, the dislocation is initialized on the (112) plane and along the  $[1\bar{1}0]$  direction, which is perpendicular to its Burgers vector of  $[11\bar{1}]$ , as shown in Fig. 1(d). These initializations allow the PFDD model to properly track the motion of both  $[11\bar{1}]$  edge and screw dislocations gliding on a (112) plane. Both edge and screw dislocations can move by advancing one Burgers vector,  $[11\bar{1}]$ , or move in the opposite direction of the Burgers vector,  $[\bar{1}\bar{1}1]$ , based on the applied shear stress to the system.

The simulation cells are made large enough to minimize effects of periodic boundary conditions and to allow for a sufficiently large dipole spacing to achieve dipole stability under no applied stress. Simulation cell sizes are chosen to allow a dipole spacing of at least  $64b$  and a distance of at least  $32b$  between each monopole and the boundary of the simulation cell. Special consideration was made, however, for the (112) edge dipole due to the large lattice spacing along the  $[1\bar{1}0]$  direction. In this case, the dipole spacing is  $128b$  and the distance between each monopole and the end of the simulation cell is  $64b$ .

The critical resolved shear stress (CRSS) (i.e., variously called lattice friction stress or Peierls stress) is determined

with a long ( $128b$ ) dislocation dipole. These calculations start with an initially straight screw or edge dislocation dipole placed within a  $128b \times 128b \times 128b$  simulation cell or a  $128b \times 128b \times 384b$  for the (110) edge case. A pure shear stress is applied such that the maximum resolved shear stress plane corresponds to the (110) habit plane for {110} glide or the (112) habit plane for {112} glide. To determine the twinning and antitwinning CRSS, a positive and negative shear was applied to the (112) habit plane to move the dislocation in both twinning and antitwinning directions, respectively. The minimum value of shear stress needed to move the entire line of the dislocation a distance  $1b$  over 5000 time steps is determined and assigned the CRSS. Increments of  $0.005\mu$  in applied stress are used to approach the minimum. These simulations were carried out at temperatures of 0, 300, 400, 500, and 600 K.

#### D. Molecular dynamics calculations

Material properties needed for calculation are the lattice constant ( $a$ ), elastic stiffness tensor, and  $\gamma_{\text{usf}}^\alpha$  as a function of temperature. These fundamental properties are obtained using molecular dynamics (MD) simulations for both pure W and W-20Re at temperatures ranging from 0 to 600 K. The Large-scale Atomic/Molecular Massively Parallel Simulator (lammps) code [54] is used to perform MD simulations, with atomic interactions described by the embedded atom method (EAM) interatomic potential developed for the W-Re alloy [55]. The lattice parameter and bulk elastic constants at different temperatures are computed using bcc supercells with 2000 atoms for W and W-20Re. The supercells are equilibrated using conjugate gradient minimization and Nose-Hoover thermo/barostat at different temperatures, while maintaining periodic boundary conditions in all directions. The equilibrated supercells for both materials are deformed according to the stress-strain method to obtain the three independent elastic constants ( $C_{11}$ ,  $C_{12}$ , and  $C_{44}$ ) [56,57].

The elastic constants ( $C_{11}$ ,  $C_{12}$ , and  $C_{44}$ ) are used to calculate the effective  $\mu$ , according to Hill [58], which is taken as an average of the Voigt effective modulus

$$\mu_v = \frac{C_{11} + 3C_{44} - C_{12}}{5} \quad (11)$$

and the Reuss effective modulus

$$\mu_r = \frac{5(C_{11} - 3C_{12})C_{44}}{3C_{11} - 3C_{12} + C_{44}}, \quad (12)$$

giving

$$\mu = \frac{\mu_v + \mu_r}{2}. \quad (13)$$

They are also used to calculate the Zener ratio  $A$ , a measure of the degree of cubic elastic anisotropy, given by

$$A = \frac{2C_{44}}{C_{11} - C_{12}}. \quad (14)$$

The GSFE for the (110)[ $\bar{1}11$ ] slip system in both materials is calculated using an orthogonal bcc simulation cell with the  $X$  axis oriented along the [ $\bar{1}11$ ] direction (Burgers vector), the  $Y$  axis along the [110] direction (slip plane normal), and the  $Z$  axis along the [ $1\bar{1}2$ ] direction (dislocation line length).

Similarly, the simulation cell for a (112)[ $11\bar{1}$ ] slip system has the  $X$ ,  $Y$ , and  $Z$  axis oriented along the [ $11\bar{1}$ ] (Burgers vector), [112] (slip plane normal), and [ $\bar{1}10$ ] (dislocation line length) direction, respectively, for both W and W-Re. The (110)[ $\bar{1}11$ ] slip system simulation cell consists of 48 atoms on each of the 30 (110) atomic planes, whereas the (112)[ $11\bar{1}$ ] slip system simulation cell has 48 (112) slip planes with 32 atoms on each plane. The simulation cells are equilibrated using conjugate gradient minimization and Nose-Hoover thermo/barostat at different temperatures. Next, a vacuum region of  $\sim 2.5$  nm is included along the slip plane normal to minimize periodic image interactions between stacking faults while maintaining periodic boundary conditions. To obtain a  $\gamma_{\text{usf}}^\alpha$  value, the upper half of the crystal is displaced along the slip direction from 0 to  $1b$  in 20 discrete shifts to produce a generalized stacking fault energy curve for each of the slip systems, where the maximum of the curve corresponds to  $\gamma_{\text{usf}}^\alpha$ . At each displacement, structural relaxation and energy minimization are performed at constant temperature with atoms allowed to relax only along the direction normal to the slip plane [59,60]. GSFE calculations are performed for the central 10 (110) and (112) slip planes of W-Re alloy to represent varying local compositions in a local neighborhood of two slip planes involved in creating the stacking fault. The  $\gamma_{\text{usf}}^\alpha$  for that atomic environment represents the barrier against glide resulting from the atomic configuration, composition, and lattice distortion for that atomic environment at finite temperature. The  $\gamma_{\text{usf}}^\alpha$  and asymmetry parameter  $B$  were averaged over the 10 central slip planes.

### III. RESULTS

#### A. Elastic properties and GSFE of W and W-20Re on {110} and {112}

The lattice and anisotropic elastic constants are calculated for W and W-20Re using the EAM W-Re YC2 potential [55] at 0, 300, 400, 500, and 600 K. The calculated elastic properties are tabulated in Table I.

The calculated elastic constants agree well with experimentally determined single-crystal elastic properties of W near room temperature [61,62]. The lattice parameter  $a$  and Burgers vector magnitude  $b$  increase with increased temperature for both W and W-20Re. Alloying with 20%Re increases  $a$  and  $b$  from those of pure W.

The results show that both elevated temperatures and Re alloying additions enhance elastic anisotropy. W is well known to be elastically isotropic and, as expected, the calculated  $A$  of pure W is unity at 0 K. However, as temperature increases,  $A$  monotonically decreases slightly below 1.0, reflecting a gradual increase in anisotropy. For W-20Re,  $A$  begins well below unity at 0 K, indicating that 20%Re induces elastic anisotropy, and continues to decrease with increasing temperature. Additionally, at all temperatures tested, 20%Re decreases  $\mu$  by about 30 GPa compared to W.

Figure 2 shows the GSFE curves for both materials at 0, 300, 400, 500, and 600 K. For W-20Re, the GSFE curve shown is an average of 10 GSFE curves with varying 80%W-20%Re configurations on the slip plane. For both W and W-20Re, the (110) curves are symmetric, and for (112), they

TABLE I. Elastic properties of W and W-20Re at different temperatures calculated using the EAM W-Re YC2 interatomic potential, used as input parameters for PFDD simulations.

| Material     | $a$ (Å) | $b$ (Å) | $C_{11}$ (GPa) | $C_{12}$ (GPa) | $C_{44}$ (GPa) | $\mu$ (GPa) | $A$  |
|--------------|---------|---------|----------------|----------------|----------------|-------------|------|
| W 0 K        | 3.165   | 2.741   | 522.1          | 204.0          | 161.1          | 160         | 1.0  |
| W 300 K      | 3.170   | 2.745   | 520.9          | 197.6          | 157.0          | 159         | 0.97 |
| W 400 K      | 3.174   | 2.749   | 520.2          | 193.8          | 154.6          | 158         | 0.95 |
| W 500 K      | 3.177   | 2.751   | 519.7          | 191.3          | 153.0          | 157         | 0.93 |
| W 600 K      | 3.180   | 2.754   | 519.0          | 187.6          | 150.7          | 157         | 0.91 |
| W-20Re 0 K   | 3.177   | 2.751   | 499.7          | 213.9          | 125.1          | 132         | 0.88 |
| W-20Re 300 K | 3.184   | 2.757   | 494.8          | 208.0          | 122.0          | 131         | 0.85 |
| W-20Re 400 K | 3.186   | 2.759   | 493.1          | 206.2          | 121.0          | 130         | 0.84 |
| W-20Re 500 K | 3.186   | 2.760   | 491.6          | 204.5          | 120.1          | 130         | 0.84 |
| W-20Re 600 K | 3.190   | 2.763   | 488.7          | 201.5          | 118.7          | 129         | 0.83 |

are asymmetric. From these GSFE curves, we identify the two key parameters needed to describe the lattice energy [Eq. (5) and Eq. (8)]: the peak value, called the unstable stacking fault energy ( $\gamma_{\text{usf}}^\alpha$ ), and the displacement at which the peak is reached. These values are summarized in Table II. For

(110), the displacement for the peak is  $0.5b$ . The  $\gamma_{\text{usf}}^\alpha$  for (112) is higher than for (110). For W, in Figs. 2(a) and 2(b), the  $\gamma_{\text{usf}}^\alpha$  difference of  $\sim 250$  mJ/m<sup>2</sup> is consistent at all temperatures. For W-20Re, the increase in  $\gamma_{\text{usf}}^{(112)}$  from  $\gamma_{\text{usf}}^{(110)}$  is less, about 200 mJ/m<sup>2</sup>, for all temperatures. The pure W at 0 K

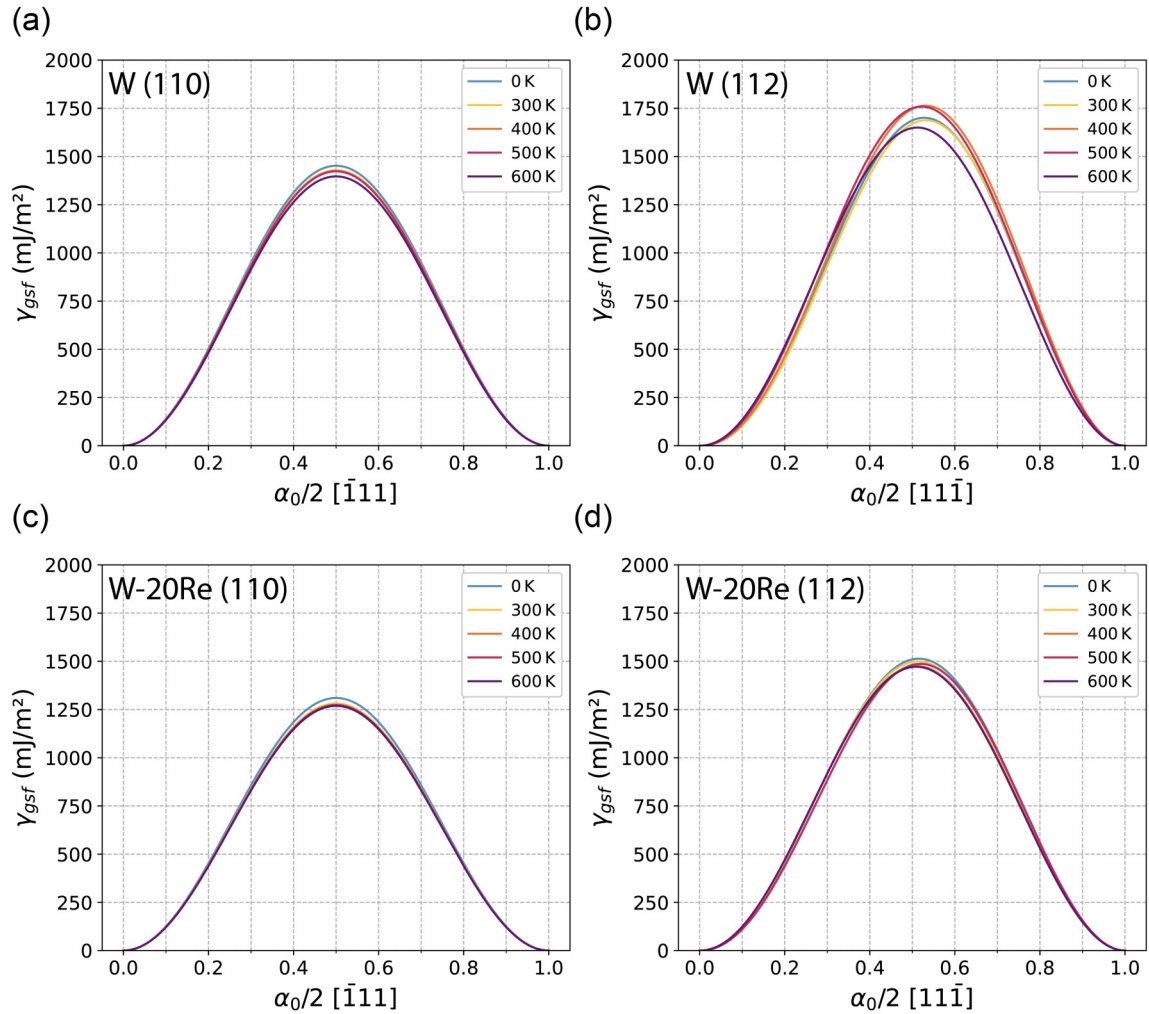


FIG. 2. Generalized stacking fault curves calculated using the W-Re YC2 interatomic potential for W for the (a) (110) plane and (b) (112) plane, and the generalized stacking fault curves for W-20Re (averaged from 10 realizations) for the (c) (110) plane and (d) (112) plane at 0, 300, 400, 500, and 600 K.

TABLE II. Parameters used to describe  $\psi_{\text{latt}}$  for {110} and {112} in W, W-20Re calculated with EAM W-Re-YC2.

| Material     | $\gamma_{\text{usf}}^{(110)}$ (mJ/m <sup>2</sup> ) | $\delta_{\text{peak}}^{(110)}$ ( <i>b</i> ) | $\gamma_{\text{usf}}^{(112)}$ (mJ/m <sup>2</sup> ) | $\delta_{\text{peak}}^{(112)}$ ( <i>b</i> ) |
|--------------|----------------------------------------------------|---------------------------------------------|----------------------------------------------------|---------------------------------------------|
| W 0 K        | 1450                                               | 0.5                                         | 1700                                               | 0.53                                        |
| W 300 K      | 1430                                               | 0.5                                         | 1690                                               | 0.53                                        |
| W 400 K      | 1420                                               | 0.5                                         | 1760                                               | 0.53                                        |
| W 500 K      | 1420                                               | 0.5                                         | 1760                                               | 0.52                                        |
| W 600 K      | 1400                                               | 0.5                                         | 1650                                               | 0.51                                        |
| W-20Re 0 K   | 1310                                               | 0.5                                         | 1510                                               | 0.52                                        |
| W-20Re 300 K | 1280                                               | 0.5                                         | 1500                                               | 0.51                                        |
| W-20Re 400 K | 1280                                               | 0.5                                         | 1480                                               | 0.51                                        |
| W-20Re 500 K | 1270                                               | 0.5                                         | 1490                                               | 0.52                                        |
| W-20Re 600 K | 1270                                               | 0.5                                         | 1470                                               | 0.51                                        |

exhibits a slightly lower  $\gamma_{\text{usf}}^{\alpha}$  than those reported for both {110} (around 1750 mJ/m<sup>2</sup>) and {112} (around 1850 mJ/m<sup>2</sup>) [49]. Although the  $\gamma_{\text{usf}}^{\alpha}$  is not the only contribution to the CRSS, it does indicate that we can expect the CRSS for {112} to be higher than {110} and temperature not to change this ranking.

We observe a small reduction in  $\gamma_{\text{usf}}^{\alpha}$  as the temperature increases, with one exception: the  $\gamma_{\text{usf}}^{(112)}$  for pure W becomes slightly elevated at 400 and 500 K. No finite-temperature {112} GSFE curves for pure W have been reported previously. Here, the (112)[11 $\bar{1}$ ] GSFE calculation was repeated for W using a W-Ta interatomic potential [63] where the  $\gamma_{\text{usf}}^{(112)}$  continually decreased with temperature. The difference only highlights that temperature dependency, although small, is sensitive to the interatomic potential that is used. Nonetheless, as our interest lies in elucidating Re effects, all atomic-scale inputs are consistently taken from simulation using the same EAM W-Re-YC2 potential.

Regarding Re effects, two observations are noteworthy. Compared to pure W, the  $\gamma_{\text{usf}}^{\alpha}$  for W-20Re is lower by  $\sim 150$ – $200$  mJ/m<sup>2</sup> for both (110) and (112) for all tested temperatures. Re alloy additions have lowered the energy barrier for fault creation. For both W and W-20Re, as the temperature increases, the (112) GSFE curves become less asymmetric, with the  $\gamma_{\text{usf}}^{\alpha}$  peak location closer to  $0.5b$  and the asymmetry parameter  $B$  approaching unity. For each temperature, the (112) GSFE for pure W shows a greater degree of asymmetry compared to the (112) GSFE for W-20Re. This difference could translate to a greater difference in CRSS between twinning and antitwinning directions on {112} for pure W.

### B. Comparing PFDD and anisotropic antiplane solution displacement fields

To verify the PFDD extension to {112} slip, we compare the PFDD displacement fields with the linear elastic antiplane solution for a straight {111} screw dislocation in a cubic anisotropic continuum [6]. The out-of-plane displacement  $u_3$  is given by

$$u_3 = \frac{b}{2\pi} \arctan 2(\zeta, \eta), \quad \text{with} \quad \begin{bmatrix} \zeta \\ \eta \end{bmatrix} = \mathcal{H} \begin{bmatrix} x \\ y \end{bmatrix}, \quad (15)$$

where  $\mathcal{H}$  is a  $2 \times 2$  anisotropy matrix representing the linear stretch and rotation of the in-plane coordinates.

Figure 3 shows the screw line direction displacement fields,  $u_{[111]}$  for the (110) screw and  $u_{[11\bar{1}]}$  for the (112) screw, as well as the differences between the displacements calculated by the linear elastic antiplane solution and those predicted from the PFDD model. Each field is normalized by its own  $|u_{\text{max}}|$ . Apart from the discretized nature of the PFDD fields, the normalized PFDD displacement fields closely match the analytic solution. A residual analysis gives root mean square error (RMSE)  $\sim 0.13$  for {110} and RMSE  $\sim 0.22$  for {112}, with the largest discrepancies confined to the core region where the solution does not resolve a core structure. Away from the core, differences are only a few percent ( $\sim 4\%$  for {110} and  $\sim 6\%$  for {112}). These comparisons confirm that the {112} implementation reproduces the expected antiplane character, angular variation, and far-field decay of an  $\frac{a}{2}$ {111} screw with a level of accuracy comparable to that for {110}.

### C. Screw and edge dislocation structures on {110} and {112}

To characterize the equilibrium core structure of dislocations on {110} and {112} planes, screw and edge dislocations in pure W were initialized on the (110) and (112) habit plane and allowed to relax under zero applied stress at 0 K until convergence. The final order parameter fields surrounding the dislocation core are shown in Fig. 4. W and W-20Re show similar core structures for screw and edge dislocations. The corresponding order parameter fields for W-20Re are provided in Fig. 11 of the Appendix. For both (110) and (112) screw dislocations, the core extends onto the cross-slip planes, indicating a nonplanar core. For the (110) screw dislocations, the distribution is symmetric, whereas for the (112), spreading on the cross-slip planes is asymmetric. For the edge dislocations, the core is planar.

### D. CRSS calculations at finite temperature

The CRSS to initiate glide can be related to material strength. Figures 5 and 6 compare the CRSS, using two different normalizations, between W and W-20Re for a range of temperatures for (a) (110) dislocations and (b), (c) (112) dislocations. All CRSS values are tabulated in Table IV of the Appendix. Regardless of normalization, some general trends emerge. The CRSS for screw dislocations exceeds that of edge dislocations at all temperatures and glide planes. At any given temperature, the CRSS remains significantly higher on (112) than on (110) for both edge and screw dislocations. Therefore, neither higher temperatures nor Re additions make {112}

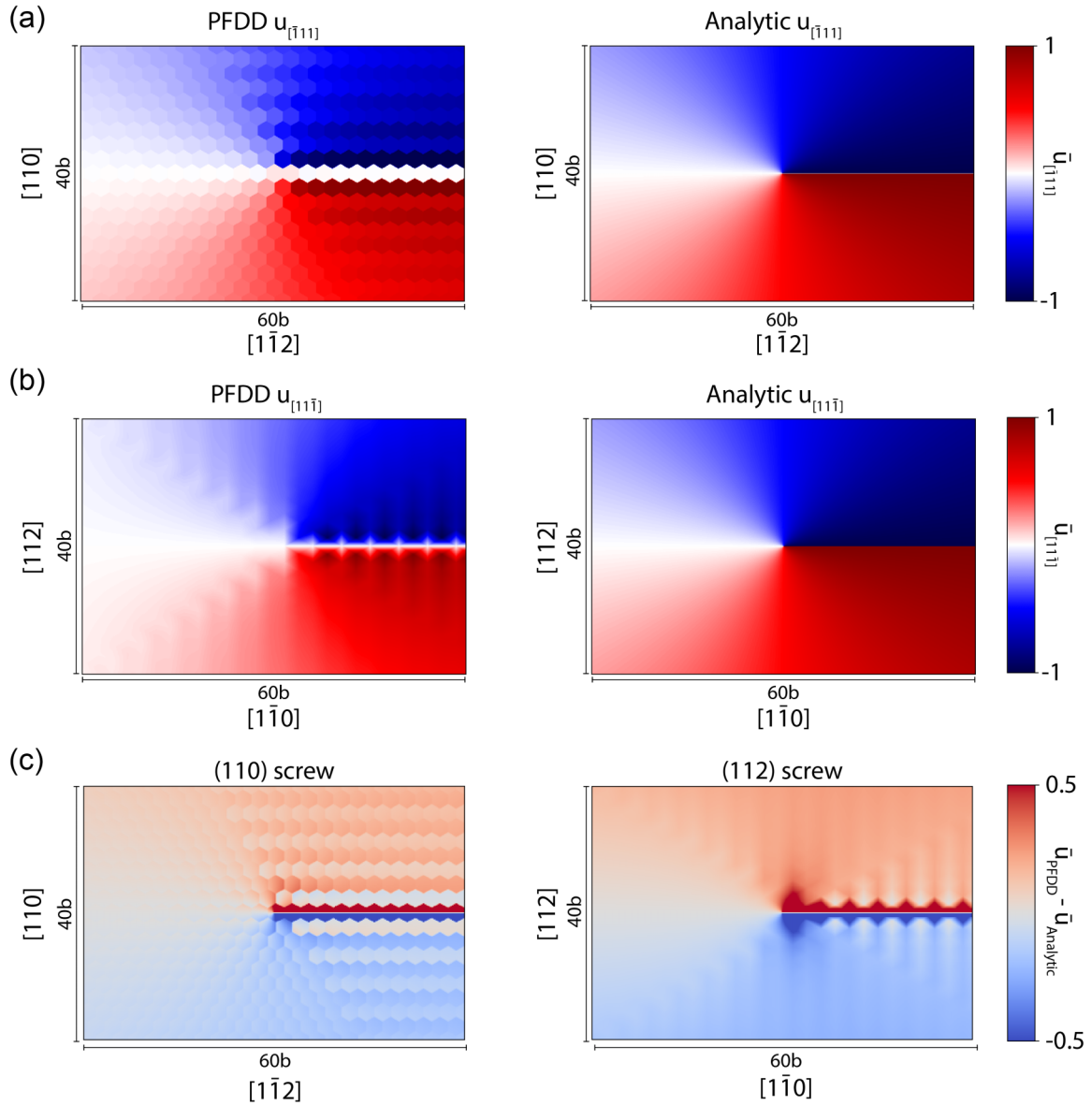


FIG. 3. Displacement fields calculated from PFDD and the elastic cubic anisotropic antiplane solution for initialized screw dislocations in pure W at 0 K and zero applied stress on the (a) (110) plane, (b) (112) plane, and (c) difference in displacement fields between PFDD and the analytical solution for (110) and (112) screw dislocations.

glide equally or easier to activate than  $\{110\}$  glide or brought the screw glide resistance closer to that of the edge glide.

To elucidate the effect of Re alloying additions and temperature, Fig. 5 analyzes the CRSS normalized by the corresponding value of  $W$  at 0 K. Differences among these normalized CRSS values will include differences in the stiffness moduli between  $W$  and  $W$ -20Re. For the same temperature, this normalized CRSS is lower for  $W$ -20Re than  $W$ , without exception. Notably, this softening Re effect is much greater for  $\{112\}$ T glide and  $\{112\}$ AT glide, for both screw and edge dislocations, than  $\{110\}$  slip. In fact, the Re effect on  $\{110\}$  glide is arguably slight, especially above 300 K.

Re reduces the elastic moduli relative to those of pure  $W$  and, for both, they decrease as temperature increases. To remove the effects of temperature on stiffness and differences between their stiffnesses, the CRSS in Fig. 6 is normalized

by the corresponding temperature-dependent isotropic effective shear modulus,  $\mu$ . For (110) dislocations, the normalized CRSS still decreases with temperature, indicating additional temperature effects on the mechanisms and kinetics of glide. In  $W$ , edge dislocations are virtually temperature insensitive in the range of 400–600 K, but for  $W$  screws and  $W$ -20Re edge and screws, the CRSS continually reduces with temperature in this same range. For temperatures above room temperature, the temperature dependence is less pronounced for  $\{112\}$  glide than for  $\{110\}$  glide. A temperature insensitive regime for the CRSS of  $\{112\}$  glide ranges from 300 to 500 K, except for screw dislocation  $\{112\}$ T glide in  $W$ -20Re, which is seen to monotonically decrease with temperature.

Figures 6(b) and 6(c) highlight the twinning/antitwining (T/AT) asymmetry in glide resistance for (112) dislocations. At 0 K, for both  $W$  and  $W$ -20Re, the CRSS for slip along the

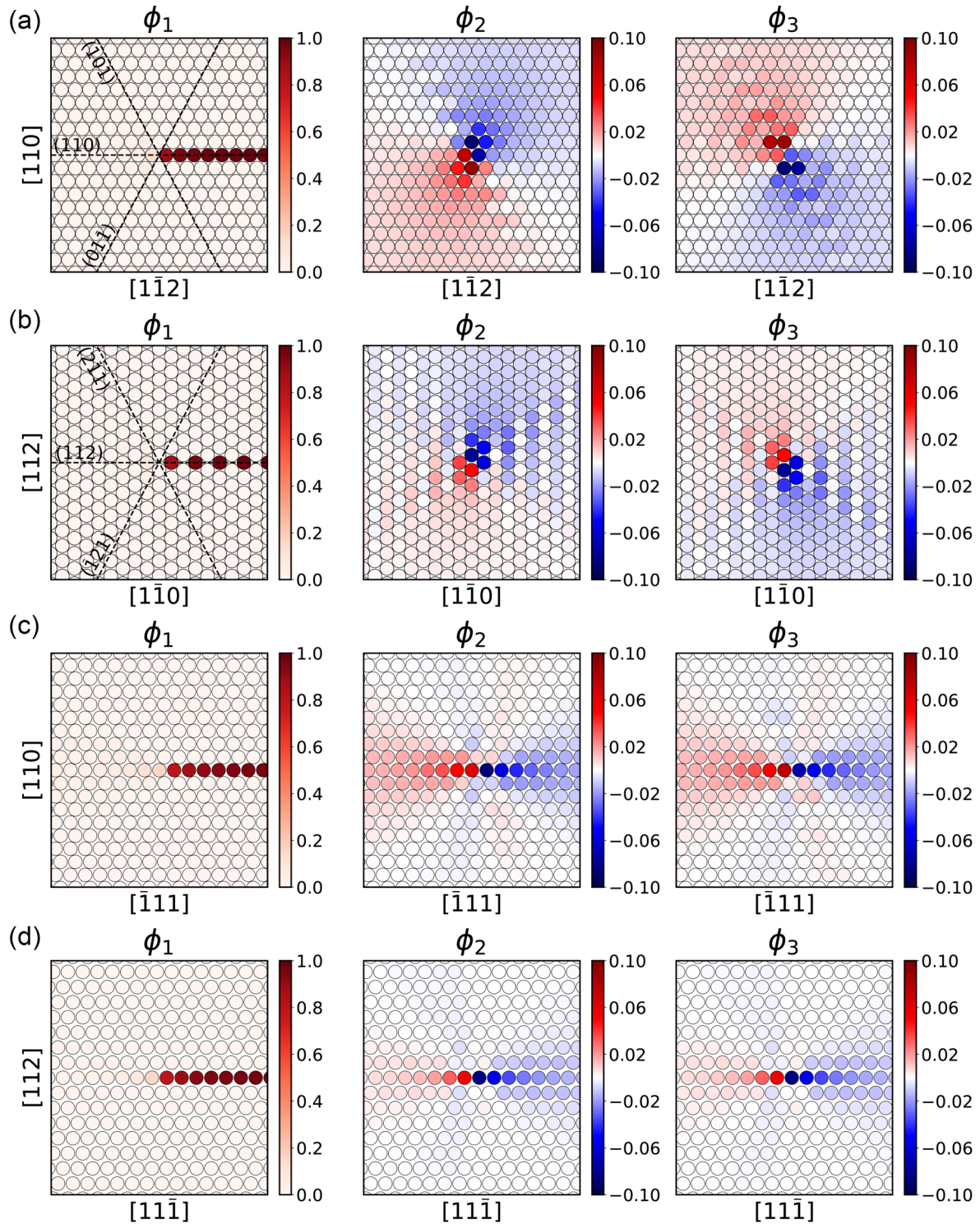


FIG. 4. Order parameter fields of relaxed dislocations in pure W at 0 K and zero applied stress for (a) (110) screw, (b) (112) screw, (c) (110) edge, and (d) (112) edge. Dashed lines for the screw cases indicate possible slip planes associated with {110} and {112}.

twinning (T) direction is less than the corresponding CRSS for slip in the antitwinning (AT) direction for both screw and edge dislocations. However, as temperature increases, the T/AT CRSS ratio approaches unity, reflecting a reduction in resistance asymmetry between the two directions and an effective increase in the number of available slip systems. In

W-20Re, the bias is removed at 400 K, while in pure W, it occurs at a higher temperature of 600 K.

#### E. Screw dislocation glide modes on {110} and {112} systems

Towards understanding thermally activated behavior and potentially ductility, we seek to understand the effect of Re

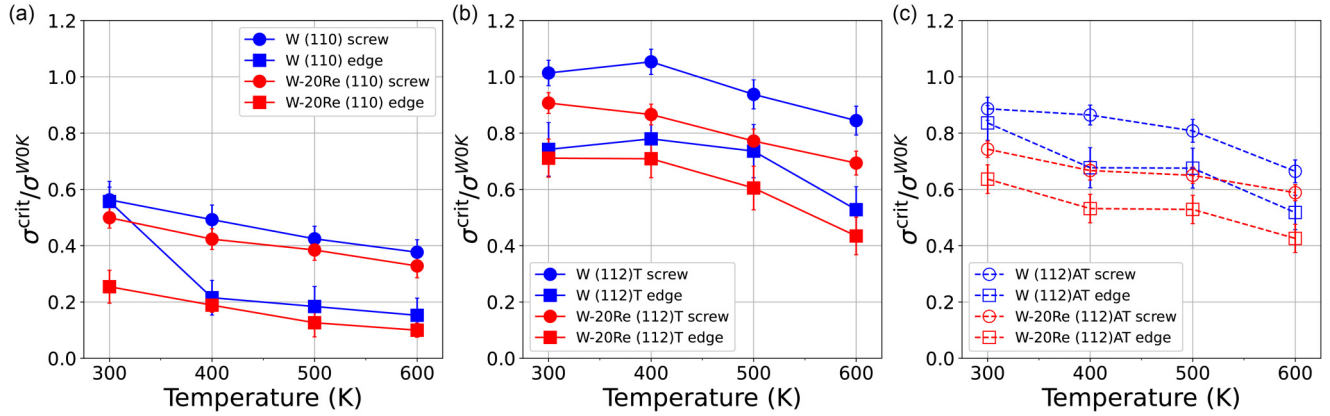


FIG. 5. Ratio of CRSS to 0 K CRSS for screw and edge dislocations in W and W-20Re as a function of temperature for (a) (110), (b) (112)T, and (c) (112)AT glide. Error bars show variation among four realizations.

on dislocation glide mechanisms. Thus far, we sought to find a critical threshold stress to activate motion, applying subcritical stress levels in fine increments until the dislocation starts to glide. To study glide mechanisms, however, a constant stress sufficiently large to sustain continuous dislocation motion across the plane is applied, which is equal to or greater than the CRSS. To compare behavior between {110} and {112} glide and for both materials, this stress or range of stresses will be the same.

The glide mechanisms are expected to vary with stress and temperature. Previous studies of pure bcc metals and alloys on {110} have shown that screw dislocations in bcc materials can exhibit many different glide mechanisms, dependent on the temperature and stress applied to the system [40,64]. In the single-kink-pair (skp) glide mode, also known as the classic kink-pair mechanism [6,65], the screw dislocation advances via the nucleation and subsequent lateral migration of one kink pair along the dislocation line [Fig. 7(a)] or widely spaced kink pairs (e.g., in this case, >30 nm). This process repeats itself for successive  $1b$  advancements. During skp motion, the dislocation line exhibits extended incubation periods before the first kink-pair nucleation event, followed by prolonged dwell times between successive nucleation events.

In contrast, the multiple-kink-pair (mkp) mode involves nucleation of many closely spaced kink pairs that interact to advance the screw dislocation line forward by  $1b$ , with significantly higher kink-pair nucleation rates and shorter dwelling times than skp [Fig. 7(a)]. At higher stresses/temperatures, an overlapping-kink-pair (okp) glide mode is observed, where kink pairs nucleate on previously formed kinks, causing the dislocation line to become wavy with local advances of  $2b$  or more relative to other segments [bottom of Fig. 7(a)]. In prior work, the temperature associated with a transition from skp to mkp in {110} glide was strongly correlated with the experimentally measured DBTT for five bcc metals [64]. The higher transition temperature from mkp to okp corresponds to the *athermal yield strength* of a bcc metal, above which yield strength appears to be temperature insensitive [64].

In the simulations here, the number of unique and distinct kps along the dislocation line and in time as the screw dislocation glides is identified and the glide mode is determined. Figure 7 shows the dominant glide mode observed in the form of maps with temperature and stress axes. The maps for (110), (112)T, and (112)AT screws in W [Fig. 7(b)] and for W-20Re [Fig. 7(c)] reveal that these ranges are sufficiently broad to witness temperature transitions from skp to mkp and from

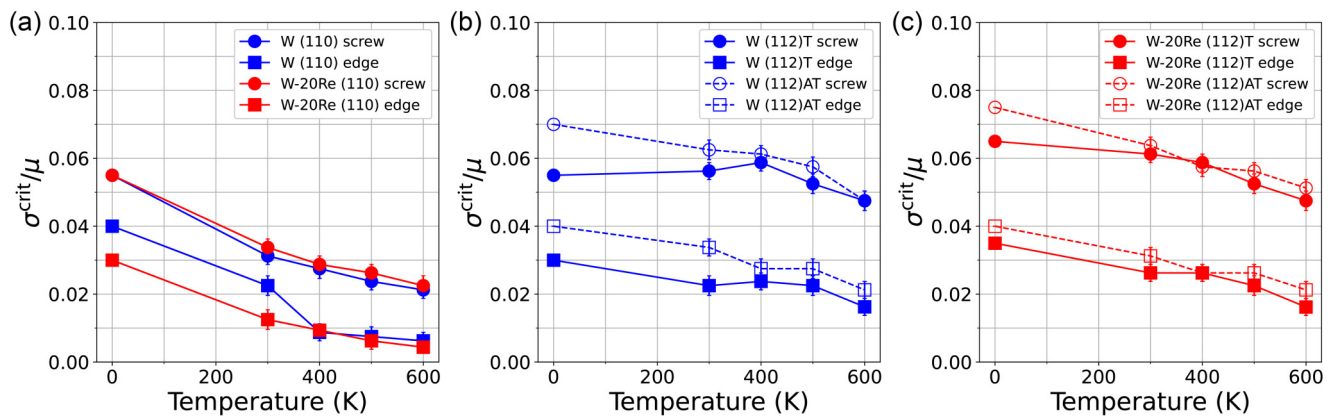


FIG. 6. Critical resolved shear stresses of screw and edge dislocations as function of temperature for (a) (110) glide in W and W-20Re, (b) (112) glide in both twinning (T) and antitwinning (AT) directions for W, and (c) in both T and AT directions for W-20Re. Error bars show variation among four realizations.

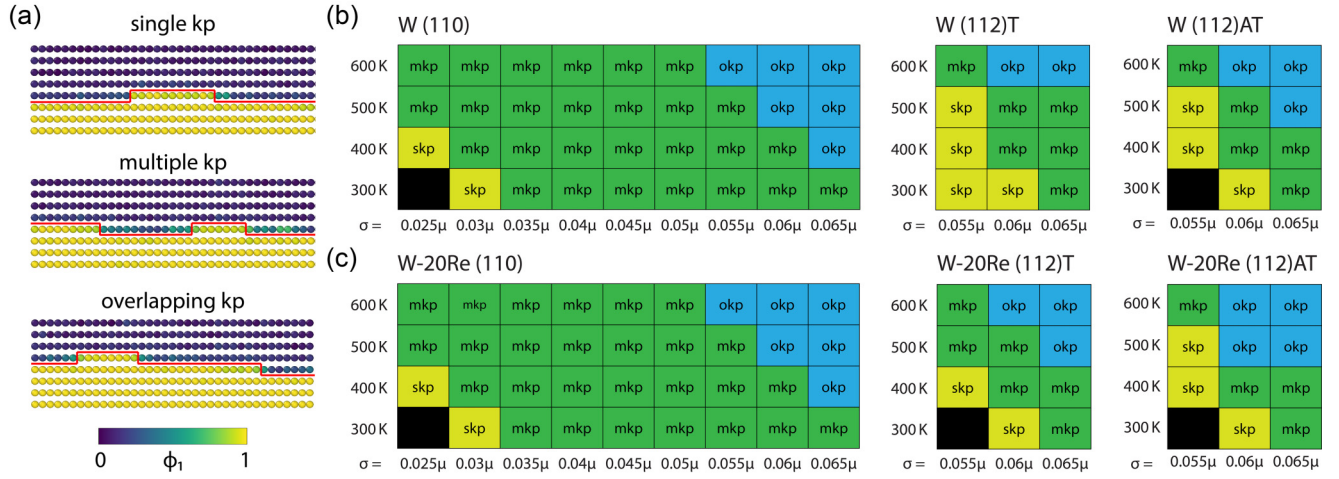


FIG. 7. Zoomed-in regions of the simulation that highlight instances of kink formation for (a) single-kink-pair (skp), multiple-kink-pair (mkp), and overlapping-kink-pair (okp) slip modes, and (b), (c) kink-pair mechanism maps as a function of applied stress and temperatures for (110), (112)T, and (112)AT screw dislocations in (b) W and (c) W-20Re. Black regions indicate no dislocation motion for the time limit of 10 000 time steps.

mkp to okp as temperature and stress increase. For {110}, the temperature and stress ranges of active glide are wider than for {112} and, over these regions, both W and W-20Re (110) screw dislocations use the same glide modes. For the range studied, the dominant mode is mkp. Re thus has little influence on the glide mechanism of {110} screw dislocations.

In contrast, Re influences the glide mode for {112} screw dislocations. The transition from skp to mkp is lower for W-20Re when compared to W. As mentioned, previous work reported a link between the skp-to-mkp transition temperature for {110} screw dislocations to the DBTT in a cache of bcc metals. In that work, {112} dislocations were not studied. It is possible that the reduction in the skp-to-mkp transition temperature for {112} screw dislocations in W-20Re could contribute to a lower DBTT than W.

#### F. Activation parameters for slip on {110} and {112}

Earlier, we analyzed the threshold stress (CRSS) to activate glide, which arguably is related to the elastic-plastic transition or the yield stress. Once activated, glide mechanisms and the rate of glide could draw a closer link to ductility. Provided that the applied stress is greater than the critical stress at a given temperature, the screw dislocation glides at a steady velocity. For each dislocation in the dipole, its glide rate  $r(\xi, T)$  under constant normalized stress,  $\xi = \sigma/\mu$ , is determined from a linear regression of average dislocation position along the line versus time. In the event this rate is determined by thermally activated processes, it could be related to stress and temperature via an Arrhenius relationship, involving an activation barrier, i.e.,

$$r(\xi, T) = v(\xi, T)e^{-\frac{\Delta H^{\text{eff}}}{k_B T}}, \quad (16)$$

where the prefactor  $v(\xi, T)$  is generalized to be stress and temperature dependent,  $\Delta H^{\text{eff}}$  is the effective activation enthalpy barrier, and  $k_B$  is Boltzmann's constant. Figure 8 shows  $\ln r$  versus  $1/T$  for W and W-20Re for (110), (112)T, and (112)AT screw dislocations under  $\xi = 0.055$ , over the

temperature range 300–600 K. At this stress level and for all temperatures, glide rates on {110} are higher than on {112}T, which are both greater than {112}AT. Further,  $\ln(r)$  scales linearly with  $1/T$ , suggesting  $r(\xi, T)$  can be described by Eq. (16) and its slope corresponds to an effective  $\Delta H^{\text{eff}}$ . From Fig. 7, we observe that over this range of  $T$ , the underlying glide mechanism changes with  $T$  and hence  $\Delta H^{\text{eff}}$  barrier would not be associated with a specific glide mechanism and  $v(\xi, T)$  possibly includes temperature and stress dependencies.

The screw dislocation glide rate analysis was repeated for a range of  $\xi = 0.045$  to  $0.065$ . The ranking in glide rates seen earlier for  $\xi = 0.055$ , i.e., {110} > {112}T > {112}AT, is preserved. In all cases,  $\ln(r)$  and  $1/T$  are linearly related and an  $\Delta H^{\text{eff}}$  could be determined.

Figure 9 shows the  $\Delta H^{\text{eff}}$  for (110), (112)T, and (112)AT screw dislocations with increasing  $\xi$ . For a given material,  $\Delta H^{\text{eff}}$  for a (110) screw dislocation is an order of magnitude lower than that of a (112)T and (112)AT screw. Also, for both materials and all stress states, the expected AT/T asymmetry on (112) is maintained, i.e.,  $\Delta H^{\text{eff}}_{\text{AT}} > \Delta H^{\text{eff}}_T$ . Last, for all temperatures and stresses and with and without Re, (110) screws glide much faster than (112) screws. However, there are two effects of Re that are noteworthy. Re alloying slightly raises  $\Delta H^{\text{eff}}$  above that of W in the low-stress regime, which would suggest that Re additions could lower {110} dislocation mobility (at least over the temperature range 300–600 K). In contrast, Re significantly reduces  $\Delta H^{\text{eff}}$  for the (112)T and (112)AT dislocations. Evidently, Re strongly impacts {112}T/AT glide and, compared to {110}, has the opposite effect of enhancing {112} mobility. Put another way, as better seen in the compiled data in Fig. 9(d), Re reduces the gap in  $\Delta H^{\text{eff}}$  between the {110} and {112} glide modes.

Figure 9(d) also highlights a great sensitivity of  $\Delta H^{\text{eff}}$  to stress for (112) screws compared to (110) screws. This difference would suggest that the activation volumes for {112} are higher than {110}. The activation volume  $v^*$  is calculated for

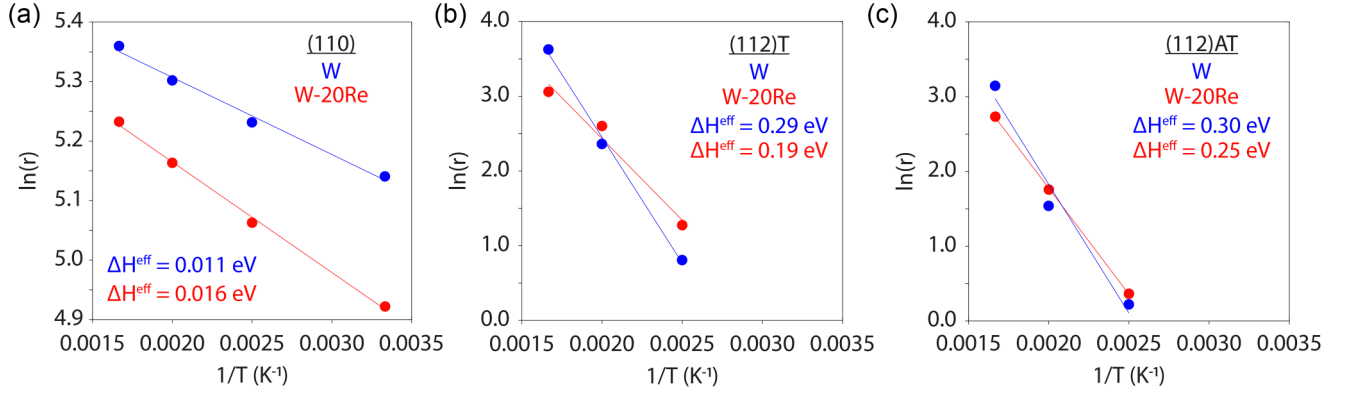


FIG. 8.  $\ln(r)$  vs  $1/T$  used to determine  $\Delta H^{\text{eff}}$  for screw dislocations in W and W-20Re on (a) (110) and (b) (112) glide in the twinning (T) direction and (c) (112) in the antitwining (AT) direction.

both materials in all three slip geometries using

$$v^* = -\left(\frac{\partial \Delta H^{\text{eff}}}{\partial \xi}\right). \quad (17)$$

The  $v^*$  are estimated from the data in Fig. 9 and summarized in Table III.

The study of  $v^*$  can provide insight into the average of many nucleation events and whether the change in slip plane

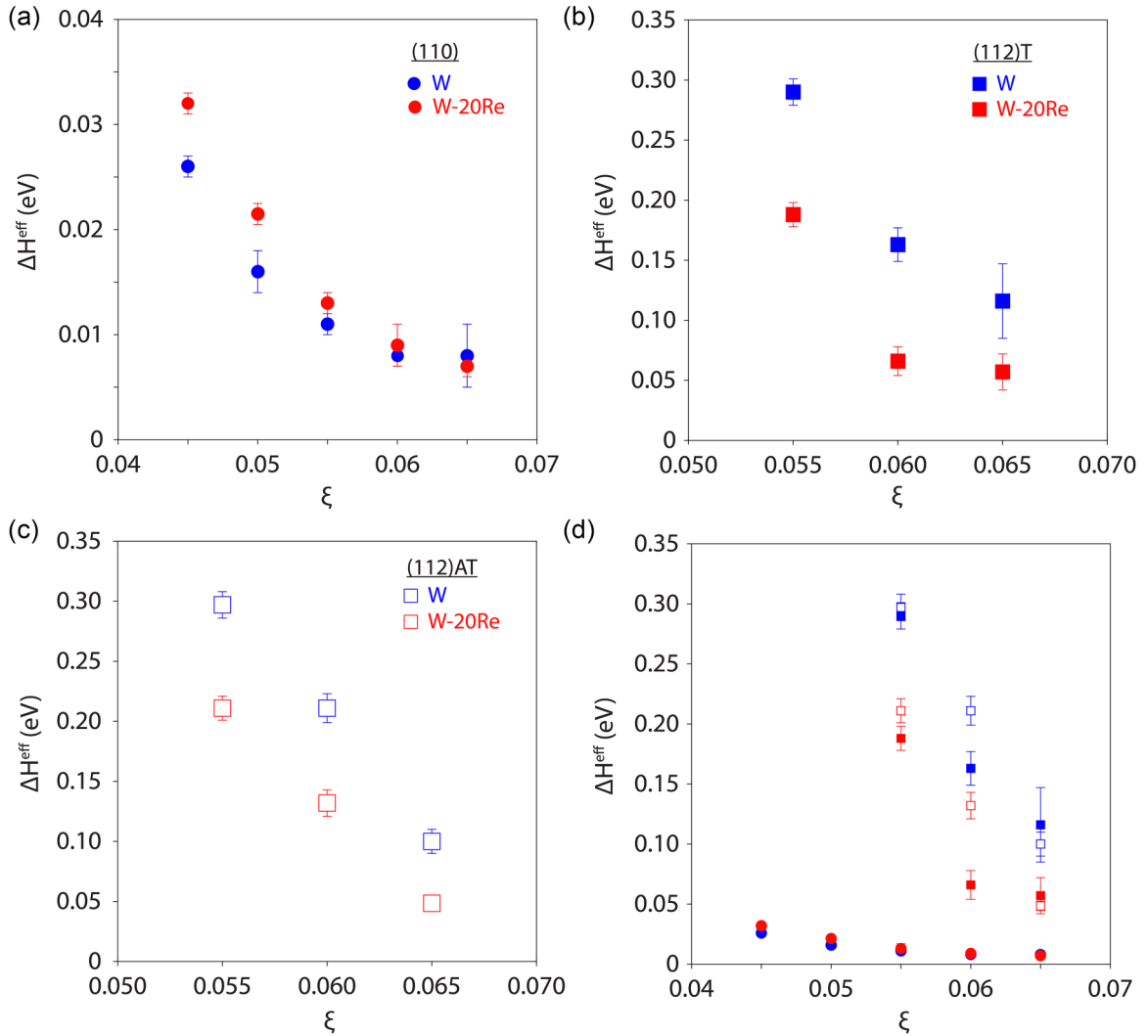


FIG. 9. Calculated  $\Delta H^{\text{eff}}$  for screw dislocations at temperatures 300–600 K as a function of  $\xi$  for (a) (110), (b) (112)T, and (c) (112)AT glide, and (d) all cases.

TABLE III. Activation volumes for (110), (112)T, and (112)AT screw dislocations in W and W-20Re over a temperature range of 300–600 K.

| Material | Slip plane | $v^*(b^3)$     |
|----------|------------|----------------|
| W        | (110)      | $0.87 \pm 0.2$ |
| W-20Re   | (110)      | $1.23 \pm 0.2$ |
| W        | (112)T     | $16.9 \pm 3.8$ |
| W-20Re   | (112)T     | $13.1 \pm 6.2$ |
| W        | (112)AT    | $19.7 \pm 1.4$ |
| W-20Re   | (112)AT    | $17.9 \pm 1.7$ |

or Re additions alter the nature of the thermally activated event. The  $v^*$  for W and W-20Re are of the order of  $1b^3$  on the (110) habit plane, which is consistent with previous PFDD simulations of {110} slip in other bcc metals [50]. The  $v^*$  for the (112) slip geometries are significantly larger than those for (110), with the (112)AT geometry exhibiting higher activation volumes for both materials when compared to (112)T. On {110}, screw dislocations are expected to form more isolated and smaller kinks compared to {112}, highlighted by the reduced activation volume and extended region of mkp (Fig. 7). On {112},  $v^*$  is much larger, suggesting kink pairs to be much broader or closely interacting with nearby kink pairs. The effect of Re on  $v^*$  is minor and within the error of the estimate.

#### IV. DISCUSSION

##### A. Competition between {110} and {112} glide in W and W-20Re

The present results offer insight into the key question of Re ductilization of W from the single dislocation perspective. First, there are many characteristics of W dislocations that are unaffected by additions of Re. Calculations of CRSS would suggest that the {110} mode remains the easier glide mode and screw dislocations are harder to activate than edge, even at elevated temperatures. Therefore, we find that Re additions will not cause the {112} mode to replace the {110} mode or “fcc-like” behavior where edge and screw dislocations have similar glide resistances.

Does Re causes strengthening or softening? One hypothesis previously put forth is that Re additions in W results in solid solution softening by enhancing the mobility of  $\frac{a}{2}\langle 111 \rangle$  screw dislocations [27,28]. For activating slip, we observe a decrease in CRSS for both {110} and {112}T/AT glide with Re addition, which would support a softening effect. However, the notable prediction is that the drop is significant for the {112} mode for all temperatures and only slight for the {110} mode at low temperatures and is nearly removed at elevated temperatures. Although {112} is less likely to be activated, the reduction in its CRSS by Re addition could translate to lower strength for W-20Re than W, which would be consistent with experiments [15,17]. In active glide, Re substantially lowers the activation enthalpy for screw dislocations, but again only on the {112} planes. Nonetheless, this significant drop would enhance mobility and suggest another softening effect. Therefore, the findings here support a Re-softening effect, as opposed to a strengthening one, but perhaps unexpectedly, the effect is seen primarily for the {112} mode.

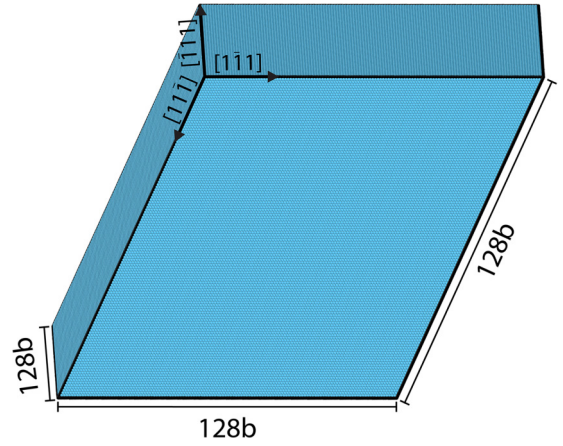


FIG. 10. The full  $128b \times 128b \times 128b$  nonorthogonal bcc simulation cell used for the PFDD simulations with primitive vectors corresponding to  $[1\bar{1}1]$ ,  $[\bar{1}11]$ , and  $[11\bar{1}]$  directions.

Others have proposed that the role of Re in W results in a shift in the primary slip mode from {110} to {112} [15,29–31]. The results here would not support this proposition. While Re makes the {112} mode easier to activate and easier to move than without, the {112} mode remains a secondary slip mode to the {110} mode. It is worthwhile to point out that Re lowers the temperature at which the glide resistances in the {112}T and AT slip directions become equivalent. Generally, having more slip systems available, that is, 12 {110} systems, 12 {112}T systems, and 12 {112}AT systems, with closer activation energies and glide resistances, will make a metal more ductile.

Overall, the results show that the effects of Re additions on the frequently overlooked secondary {112} glide mode are strong and could lead to a more ductile material. It has recently been shown that the DBTT of a refractory metal correlates with a skp-to-mkp transition temperature for the {110} mode [64]. Over a broad range of relevant stresses and temperatures, W-20Re shows a reduction in the transition temperature from skp to mkp glide on {112}, which is not seen on {110}. This result implies that Re reduces the DBTT by lowering this same transition temperature, but for the {112} mode.

##### B. Final remarks

The present model treats the W-20Re alloy as a homogeneous medium, consequently capturing the average effect of solute on the energetics of the lattice and the average changes to CRSS, activation enthalpies, and core planarity. The relatively high solute concentration of 20% Re represents a system where averaged effects of Re are most applicable. The findings enable testing ideas on the basic role of temperature and Re additions on slip mode selection between {110} and {112}T/AT. It is recognized, however, that for dilute solute additions, accounting for explicit, discrete solute-dislocation interactions would be important to resolve spatial heterogeneity in local glide resistance due to solute clustering, short-range ordering, and Cottrell atmospheres. Consequences include changes in local glide activation

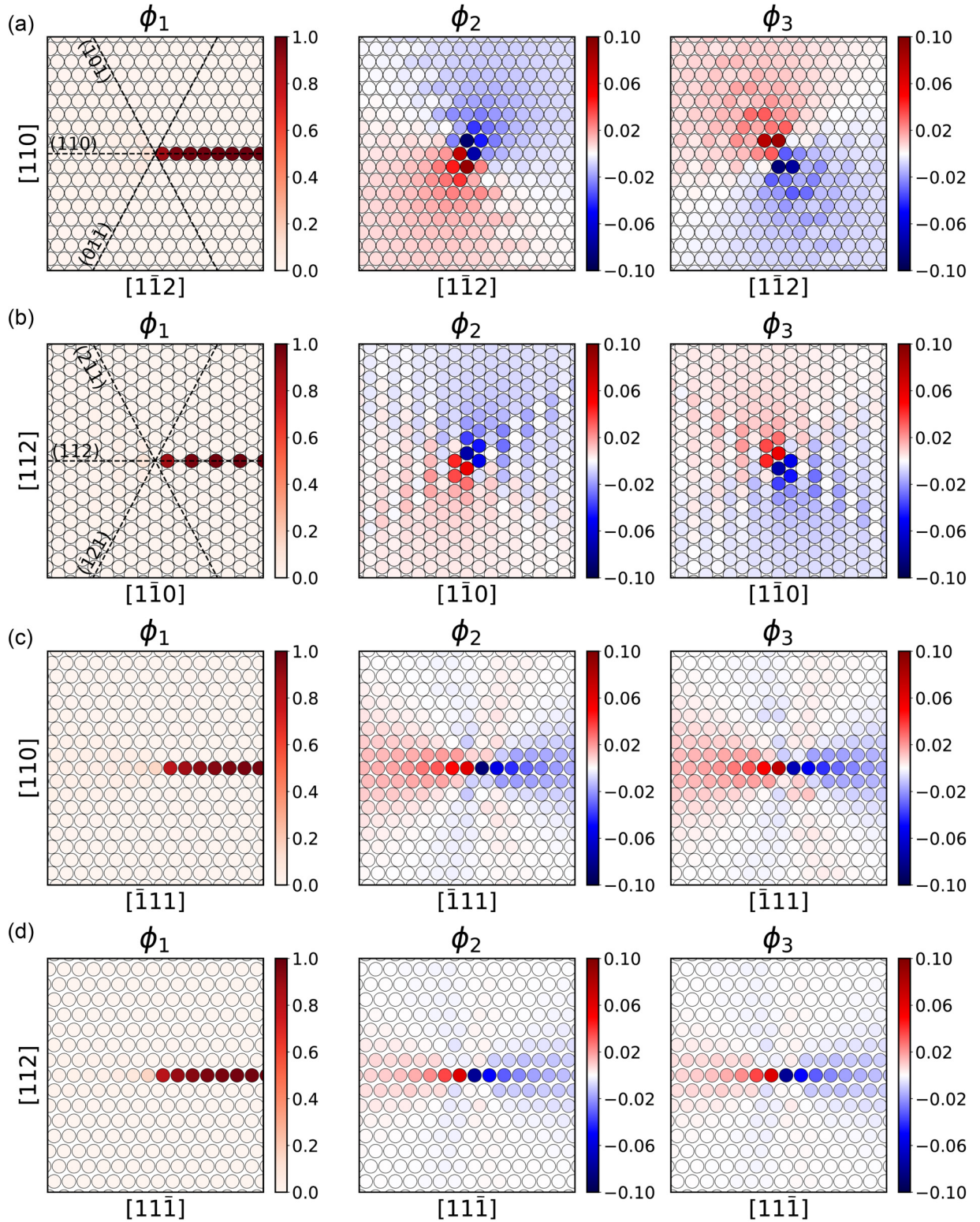


FIG. 11. Order parameter fields of relaxed dislocations in W-20Re at 0 K and zero applied stress for (a) (110) screw, (b) (112) screw, (c) (110) edge, and (d) (112) edge. Dashed lines for the screw cases indicate possible slip planes associated with  $\{110\}$  and  $\{112\}$ .

barriers, kink nucleation rates, cross-slip events, and possible pinning. These effects would add an element of stochasticity to the analysis and are possible within the current PFDD framework [66,67].

Finally, we note the anomalous increase in  $\gamma_{\text{usf}}^{(112)}$  with temperature that was observed at 400 and 500 K in pure W, contrasting with the expected behavior and the trends observed for  $\{110\}$ . Further investigation is needed to determine

TABLE IV. Averaged CRSS for screw and edge dislocations for slip on the {110} habit plane and on the {112} habit plane for twinning (T) and antitwinning (AT) directions in W and W-20Re. Values are normalized by the effective isotropic shear modulus ( $\mu$ ), which is material and temperature dependent. The values of  $\mu$  are found in Table I.

| Material     | $\sigma_{(110)\text{screw}}$ | $\sigma_{(110)\text{edge}}$ | $\sigma_{(112)\text{T}screw}$ | $\sigma_{(112)\text{T}edge}$ | $\sigma_{(112)\text{AT}screw}$ | $\sigma_{(112)\text{AT}edge}$ |
|--------------|------------------------------|-----------------------------|-------------------------------|------------------------------|--------------------------------|-------------------------------|
| W 0 K        | $0.055\mu$                   | $0.040\mu$                  | $0.055\mu$                    | $0.030\mu$                   | $0.070\mu$                     | $0.040\mu$                    |
| W 300 K      | $0.031\mu$                   | $0.023\mu$                  | $0.056\mu$                    | $0.023\mu$                   | $0.063\mu$                     | $0.034\mu$                    |
| W 400 K      | $0.028\mu$                   | $0.0088\mu$                 | $0.059\mu$                    | $0.024\mu$                   | $0.061\mu$                     | $0.028\mu$                    |
| W 500 K      | $0.024\mu$                   | $0.0075\mu$                 | $0.053\mu$                    | $0.023\mu$                   | $0.058\mu$                     | $0.028\mu$                    |
| W 600 K      | $0.021\mu$                   | $0.0063\mu$                 | $0.048\mu$                    | $0.016\mu$                   | $0.048\mu$                     | $0.021\mu$                    |
| W-20Re 0 K   | $0.055\mu$                   | $0.030\mu$                  | $0.065\mu$                    | $0.035\mu$                   | $0.075\mu$                     | $0.040\mu$                    |
| W-20Re 300 K | $0.034\mu$                   | $0.013\mu$                  | $0.061\mu$                    | $0.026\mu$                   | $0.064\mu$                     | $0.031\mu$                    |
| W-20Re 400 K | $0.029\mu$                   | $0.094\mu$                  | $0.059\mu$                    | $0.026\mu$                   | $0.058\mu$                     | $0.026\mu$                    |
| W-20Re 500 K | $0.026\mu$                   | $0.0063\mu$                 | $0.053\mu$                    | $0.023\mu$                   | $0.056\mu$                     | $0.026\mu$                    |
| W-20Re 600 K | $0.023\mu$                   | $0.0044\mu$                 | $0.048\mu$                    | $0.016\mu$                   | $0.051\mu$                     | $0.021\mu$                    |

whether this behavior is physically meaningful or a limitation of the interatomic potential. Nevertheless, this anomaly does not impact the conclusions regarding Re alloying effects presented in this work and the study of {110} and {112} slip in W and W-20Re.

## V. CONCLUSIONS

In this work, we investigate how Re influences the thermally activated glide of screw and edge dislocations in W for a range of elevated temperatures (300 to 600 K). A phase field dislocation dynamics framework is used to predict 3D dislocation line motion in a body-centered-cubic lattice and associated stresses and energetics at finite temperature. The formulation is extended here to include a second {112}{111} slip mode in addition to the {110}{111} slip mode. The only inputs required are elastic moduli and stacking fault energy surfaces from atomistic simulations. As verification, the model reproduces the distinct twinning and antitwinning asymmetry characteristic of {112}{111} glide for W and a W-Re alloy. We show that there are some room-temperature dislocation glide properties of W that higher temperatures or Re additions do not change. For both W and W-20Re and all temperatures studied, the critical resolved shear stresses (CRSS) for both screw and edge dislocations are significantly higher for {112} glide than {110} glide. The CRSS for screw dislocations substantially exceed those of edge dislocations in all cases. The key finding here is that Re affects {112} glide moreso than {110} glide. While Re additions reduce the CRSS for both {110} and {112} glide, aligned with prevailing notions that Re induces a “softening effect,” this Re softening is more significant for {112}T/AT glide, for both screw and edge dislocations, than {110} glide. Re lowers the critical temperature at which the CRSS for {112}T glide and {112}AT glide become equal in W. In the kinetics of active glide, Re substantially lowers the activation enthalpy for {112}, but slightly increases it for {110}. Re reduces the critical temperature at which the {112} glide mechanism changes from the classic kink pair to multikink pair, which has previously been associated with the ductile-to-brittle transition. These findings altogether suggest that Re alloying additions in W encourage enhanced slip activity from {110} only to both {110} and {112} modes. Accessibility of both modes

doubles the number of available slip systems (from 12 to 24) or increases the number of slip planes from 6 to 18.

## ACKNOWLEDGMENTS

P.F.M. would like to thank J. Mayer and L. Erich for helpful discussions during the duration of the project. P.F.M. and I.J.B. gratefully acknowledge support from the Department of Defense Office of Naval Research under Contract No. N00014-26-1-2021. M.R.J. was supported by the Laboratory Directed Research and Development (LDRD) program at Sandia National Laboratories. Use was made of computational facilities purchased with funds from the National Science Foundation (Grant No. CNS-1725797) and administered by the Center for Scientific Computing (CSC). The CSC is supported by the California NanoSystems Institute and the Materials Research Science and Engineering Center (MRSEC; NSF DMR No. 2308708) at UC Santa Barbara. This work was produced in part by Sandia National Laboratories, a multiprogram laboratory managed and operated by National Technology and Engineering Solutions of Sandia, LLC, a wholly owned subsidiary of Honeywell International, Inc., for the U.S. Department of Energy’s National Nuclear Security Administration under Contract No. DE-NA0003525. This paper describes objective technical results and analysis. Any subjective views or opinions that might be expressed in the paper do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

P.F.M.: methodology, data curation, data analysis, investigation, formal analysis, and writing (original draft). P.G.: data curation, investigation, formal analysis, writing (review and editing). M.R.J.: formal analysis, writing (review and editing). T.M.P.: conceptualization, writing (review and editing), supervision. I.J.B.: conceptualization, writing (review and editing), supervision, and funding acquisition.

The authors declare no competing interests.

## DATA AVAILABILITY

There are no publicly available research data or software supporting this manuscript. Requests for further information or data should be sent to the authors.

## APPENDIX

Table IV lists the averaged CRSS values on (110) and (112) in W and W-20Re. Figure 10 shows the full nonorthogonal

bcc simulation cell used for the PFDD simulations. Figure 11 shows the order parameter fields of relaxed dislocations in W-20Re.

- [1] J. Webb, S. Gollapudi, and I. Charit, An overview of creep in tungsten and its alloys, *Intl. J. Refract. Metals Hard Mater.* **82**, 69 (2019).
- [2] C. Frey, K. M. Mullin, J. Jankowski, L. H. Mills, N. R. Philips, and T. M. Pollock, A perspective on the historical and contemporary development of refractory alloys, *Metall. Mater. Trans. A Phys. Metall. Mater. Sci.* **57**, 461 (2026).
- [3] G. A. Geach and J. E. Hughes, The alloys of rhenium with molybdenum or with tungsten and having good high temperature properties, in *Plansee Proceedings* (Pergamon Press, London, UK, 1955), pp. 245–253.
- [4] Y.-H. Zhang, E. Ma, J. Sun, and W.-Z. Han, Unveiling the intrinsic rhenium effect in Tungsten, *Acta Mater.* **264**, 119586 (2024).
- [5] W. Cai and W. D. Nix, *Imperfections in Crystalline Solids* (Cambridge University Press, Cambridge, 2016).
- [6] J. P. Hirth and J. Lothe, *Theory of Dislocations*, 2nd ed. (Wiley, New York, 1982).
- [7] C. R. Weinberger, B. L. Boyce, and C. C. Battaile, Slip planes in bcc transition metals, *Intl. Mater. Rev.* **58**, 296 (2013).
- [8] F. Wang, G. H. Balbus, S. Xu, Y. Su, J. Shin, P. F. Rottmann, K. E. Knippling, J.-C. Stinville, L. H. Mills, O. N. Senkov, I. J. Beyerlein, T. M. Pollock, and D. S. Gianola, Multiplicity of dislocation pathways in a refractory multiprincipal element alloy, *Science* (1979) **370**, 95 (2020).
- [9] L. Hollang, D. Brunner, and A. Seeger, Work hardening and flow stress of ultrapure molybdenum single crystals, *Mater. Sci. Engineer. A* **319–321**, 233 (2001).
- [10] A. Seeger, The flow stress of high-purity refractory body-centred cubic metals and its modification by atomic defects, *J. Phys. IV France* **5**, C7-45 (1995).
- [11] W. Wasserbäch, Anomalous slip in high-purity niobium and tantalum single crystals, *Phys. Stat. Sol. (A)* **147**, 417 (1995).
- [12] L. L. Hsiung, On the mechanism of anomalous slip in bcc metals, *Mater. Sci. Engineer. A* **528**, 329 (2010).
- [13] C. Marichal, H. Van Swygenhoven, S. Van Petegem, and C. Borca, {110} Slip with {112} slip traces in bcc tungsten, *Sci. Rep.* **3**, 2547 (2013).
- [14] H. R. Piehler and W. A. Backofen, A theoretical examination of the plastic properties of bcc crystals deforming by  $\langle 111 \rangle$  pencil glide, *Metall. Trans.* **2**, 249 (1971).
- [15] P. L. Raffo, Yielding and fracture in tungsten and tungsten-rhenium alloys, *J. Less Comm. Metals* **17**, 133 (1969).
- [16] G. G. Lessmann and R. E. Gold, *Weldability of Tungsten Base Alloys* (Astronuclear Laboratory, Westinghouse Electric Corp., Pittsburgh, PA, 1970).
- [17] E. M. Savitskii, M. A. Tylkina, S. I. Ipatova, and E. I. Pavlova, Properties of tungsten-rhenium alloys, *Metal Sci. Heat Treat. Metals* **2**, 483 (1960).
- [18] Y. J. Hu, M. R. Fellingner, B. G. Bulter, Y. Wang, K. A. Darling, L. J. Kecskes, D. R. Trinkle, and Z. K. Liu, Solute-induced solid-solution softening and hardening in bcc tungsten, *Acta Mater.* **141**, 304 (2017).
- [19] W. R. Witzke, The effects of composition on mechanical properties of W-4Re-Hf-C alloys, *Metall. Trans.* **5**, 499 (1974).
- [20] R. R. Vandervoort, The creep behavior of W-5 re, *Metall. Trans.* **1**, 857 (1970).
- [21] W. D. Klopp, W. R. Witzke, and P. L. Raffo, *Mechanical Properties of Dilute Tungsten-Rhenium Alloys* (National Aeronautics and Space Administration, Washington D.C., 1966).
- [22] H. P. Gao and R. H. Zee, Effects of rhenium on creep resistance in tungsten alloys, *J. Mater. Sci. Lett.* **20**, 885 (2001).
- [23] C. Ren, Z. Z. Fang, M. Koopman, B. Butler, J. Paramore, and S. Middlemas, Methods for improving ductility of tungsten - A review, *Intl. J. Refract. Metals Hard Mater.* **75**, 170 (2018).
- [24] G. G. Lessmann and R. E. Gold, *Determination of Weldability and Elevated Temperature Stability of Refractory Metal Alloys II-Long-Time Temperature Stability of Refractory Metal Alloys*, NASA CR-1608 (National Aeronautics and Space Administration, Washington, D.C., 1970).
- [25] R. L. Fleischer, Solution hardening, *Acta Metall.* **9**, 996 (1961).
- [26] S. I. Rao, C. Woodward, and B. Akdim, Solid solution softening and hardening in binary bcc alloys, *Acta Mater.* **243**, 118440 (2023).
- [27] D. R. Trinkle and C. Woodward, The chemistry of deformation: How solutes soften pure metals, *Science* **310**, 1665 (2005).
- [28] A. Sato and M. Meshii, Solid solution softening and solid solution hardening, *Acta Metall.* **21**, 753 (1973).
- [29] M. Garfinkle, *Room-Temperature Tensile Behavior of 100 Oriented Tungsten Single Crystals with Rhenium in Dilute Solid Solution*, NASA TN D-3190 (National Aeronautics and Space Administration, Washington, D.C., 1966).
- [30] L. Romaner, C. Ambrosch-Draxl, and R. Pippan, Effect of rhenium on the dislocation core structure in tungsten, *Phys. Rev. Lett.* **104**, 195503 (2010).
- [31] H. Li, S. Wurster, C. Motz, L. Romaner, C. Ambrosch-Draxl, and R. Pippan, Dislocation-core symmetry and slip planes in tungsten alloys: *Ab initio* calculations and microcantilever bending experiments, *Acta Mater.* **60**, 748 (2012).
- [32] Z. Wang, R. Wu, X. Wu, and H. Li, Effect of Re, temperature and  $\sigma$  phase on mechanical behaviors of W-Re alloys: A molecular dynamics study, *Phys. Lett. A* **556**, 130829 (2025).
- [33] Y. Zhao and J. Marian, Direct prediction of the solute softening-to-hardening transition in W-Re alloys using stochastic simulations of screw dislocation motion, *Model. Simul. Mat. Sci. Eng.* **26**, 045002 (2018).
- [34] K. Srivastava, R. Gröger, D. Weygand, and P. Gumbsch, Dislocation motion in tungsten: Atomistic input to discrete dislocation simulations, *Intl. J. Plast.* **47**, 126 (2013).
- [35] Y. Wang and J. Li, Phase field modeling of defects and deformation, *Acta Mater.* **58**, 1212 (2010).
- [36] X. Peng, A. Hunter, I. J. Beyerlein, R. A. Lebensohn, K. Dayal, and E. Martinez, Non-orthogonal computational grids for bcc simulation cell used for the PFDD simulations. Figure 11 shows the order parameter fields of relaxed dislocations in W-20Re.

- studying dislocation motion in phase field approaches, *Comput. Mater. Sci.* **200**, 110834 (2021).
- [37] N. Fuchs-Lynch, M. De Leo, P. Garg, S. Xu, N. A. Mara, and I. J. Beyerlein, 3D interface size effects on slip transfer in Ti/Nb nanolaminates, *Intl. J. Plast.* **186**, 104246 (2025).
- [38] I. Wise, A. Hunter, and I. J. Beyerlein, Cross slip of extended dislocations in face-centered cubic metals through phase-field modeling, *Model. Simul. Mat. Sci. Eng.* **33**, 055011 (2025).
- [39] P. Garg, K. Hemker, and I. J. Beyerlein, Stress and temperature dependence of screw dislocation motion in refractory multi-principal element alloys, *High Entropy Alloys Mater.* **3**, 244 (2025).
- [40] P. Garg, M. R. Jones, N. E. Peterson, A. J. Clarke, and I. J. Beyerlein, Role of temperature on screw dislocation dynamics in Ta, W, and Ta–W alloy, *J. Mater. Res. Technol.* **35**, 5521 (2025).
- [41] L. T. W. Fey, A. Hunter, and I. J. Beyerlein, Phase-field dislocation modeling of cross-slip, *J. Mater. Sci.* **57**, 10585 (2022).
- [42] C. Albrecht, I. J. Beyerlein, and M. Jones, Temperature dependent phase field dislocation dynamics model, *Eur. J. Mech. A. Solids* **100**, 104987 (2023).
- [43] M. Fukuda, T. Tabata, A. Hasegawa, S. Nogami, and T. Muroga, Strain rate dependence of tensile properties of tungsten alloys for plasma-facing components in fusion reactors, *Fusion Eng. Des.* **109–111**, 1674 (2016).
- [44] P. Lin, J. Nie, S. Cui, and Y. Lu, Molecular dynamics study on the ductile-to-brittle transition in W-Re alloy systems, *Acta Mater.* **285**, 120684 (2025).
- [45] I. J. Beyerlein and A. Hunter, Understanding dislocation mechanics at the mesoscale using phase field dislocation dynamics, *Philos. Trans. R. Soc. A* **374**, 20150166 (2016).
- [46] P. F. McNutt, M. R. Jones, P. Garg, and I. J. Beyerlein, Room temperature dislocation loop dynamics in body-centered cubic refractory multi-principal element alloys, *Comput. Mater. Sci.* **244**, 113280 (2024).
- [47] M. R. Jones, Lauren T. W. Fey, and I. J. Beyerlein, High temperature dislocation glide in the MoNbTi refractory multiprincipal element alloy, *Phys. Rev. Mater.* **8**, 013604 (2024).
- [48] J. R. Mianroodi, A. Hunter, I. J. Beyerlein, and B. Svendsen, Theoretical and computational comparison of models for dislocation dissociation and stacking fault/core formation in fcc crystals, *J. Mech. Phys. Solids* **95**, 719 (2016).
- [49] S. Xu, Y. Su, L. T. W. Smith, and I. J. Beyerlein, Frank-Read source operation in six body-centered cubic refractory metals, *J. Mech. Phys. Solids* **141**, 104017 (2020).
- [50] M. R. Jones, L. T. W. Fey, and I. J. Beyerlein, Phase-field dislocation dynamics simulations of temperature-dependent glide mechanisms in niobium, *Comput. Mater. Sci.* **232**, 112652 (2024).
- [51] D. M. Herlach, Non-equilibrium solidification of undercooled metallic melts, *Metals (Basel)* **4**, 196 (2014).
- [52] G. Taylor, Thermally-activated deformation of BCC metals and alloys, *Prog. Mater. Sci.* **36**, 29 (1992).
- [53] T. Ichitsubo and K. Tanaka, Thermal fluctuation for the time-dependent Ginzburg-Landau simulation, *Phys. Rev. E* **63**, 060101(R) (2001).
- [54] A. P. Thompson, H. M. Aktulga, R. Berger, D. S. Bolintineanu, W. M. Brown, P. S. Crozier, P. J. in 't Veld, A. Kohlmeyer, S. G. Moore, T. D. Nguyen, R. Shan, M. J. Stevens, J. Tranchida, C. Trott, and S. J. Plimpton, LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales, *Comput. Phys. Commun.* **271**, 108171 (2022).
- [55] Y. Chen, J. Fang, L. Liu, W. Hu, C. Jiang, N. Gao, H.-B. Zhou, G.-H. Lu, F. Gao, and H. Deng, The interactions between rhenium and interstitial-type defects in bulk tungsten: A combined study by molecular dynamics and molecular statics simulations, *J. Nucl. Mater.* **522**, 200 (2019).
- [56] O. H. Nielsen and R. M. Martin, Stresses in semiconductors: *Ab initio* calculations on Si, Ge, and GaAs, *Phys. Rev. B* **32**, 3792 (1985).
- [57] M. González-Díaz, P. Rodríguez-Hernández, and A. Muñoz, Elastic constants and electronic structure of beryllium chalcogenides BeS, BeSe, and BeTe from first-principles calculations, *Phys. Rev. B* **55**, 14043 (1997).
- [58] R. Hill, A theory of the yielding and plastic flow of anisotropic metals, *Proc. R. Soc. London A. Math. Phys. Sci.* **193**, 281 (1948).
- [59] X. Wang, S. Xu, W. R. Jian, X. G. Li, Y. Su, and I. J. Beyerlein, Generalized stacking fault energies and Peierls stresses in refractory body-centered cubic metals from machine learning-based interatomic potentials, *Comput. Mater. Sci.* **192**, 110364 (2021).
- [60] P. Garg, M. A. Bhatia, and K. N. Solanki, Uncovering the influence of metallic and non-metallic impurities on the ideal shear strength and ductility of Ti: An *ab-initio* study, *J. Alloys Compd.* **788**, 413 (2019).
- [61] R. A. Ayres, G. W. Shannette, and D. F. Stein, Elastic constants of tungsten–rhenium alloys from 77 to 298 °K, *J. Appl. Phys.* **46**, 1526 (1975).
- [62] R. Lowrie and A. M. Gonas, Single-Crystal Elastic Properties of Tungsten from 24 ° to 1800 °C, *J. Appl. Phys.* **38**, 4505 (1967).
- [63] Y. Chen, J. Fang, L. Liu, W. Hu, N. Gao, F. Gao, and H. Deng, Development of the interatomic potentials for W-Ta system, *Comput. Mater. Sci.* **163**, 91 (2019).
- [64] M. R. Jones, P. Garg, and I. J. Beyerlein, Critical temperatures for ductile to brittle transition and athermal strength in refractory metals, *J. Mater. Res. Technol.* **38**, 6229 (2025).
- [65] A. Seeger and C. Wüthrich, Dislocation relaxation processes in body-centred cubic metals, *Il Nuovo Cimento B (1971–1996)* **33**, 38 (1976).
- [66] L. T. W. Fey, C. Reynolds, A. Hunter, and I. J. Beyerlein, Phase-field modeling of dislocation–interstitial interactions, *J. Mech. Phys. Solids* **179**, 105370 (2023).
- [67] L. T. W. Fey, A. Hunter, and I. J. Beyerlein, Role of diffusing interstitials on dislocation glide in refractory body centered cubic metals, *Model. Simul. Mat. Sci. Eng.* **32**, 045001 (2024).